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Training and Redeployment of healthcare professionals to Intensive Care during the COVID-19 pandemic: a systematic review

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Training and Redeployment of healthcare professionals to Intensive Care during the COVID-19 pandemic: a systematic review

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Abstract

Objectives: The COVID-19 pandemic has resulted in a rapid influx of patients to intensive care at a rate which exceeded pre-existing staff capacity. Resultantly, this required the rapid development of innovative redeployment and training strategies, which not only considered patient care but also infection control. The aim of this systematic review was to provide a detailed understanding of redeployment and training during the first year of the COVID-19 pandemic by capturing and considering the merit of the strategies enlisted and the experiences and needs of redeployed staff.

Design: The review involved a systematic search of key terms related to intensive care AND training AND redeployment AND healthcare workers in 9 databases. Analysis consisted of a narrative synthesis of quantitative study outputs, and a framework-based thematic analysis of qualitative study outputs and grey literature. These results were then combined applying an interpretative synthesis. We followed PRISMA and the review protocol was available online.

Results: Twenty papers were analysed. These took place primarily in the UK (N=8, 40%) and USA (N=5, 25%). Themes presented in the results are *Redeployment: Implementation strategies and learnings; Redeployed staff experience and strategies to address their needs; Redeployed staff learning needs; Training formats offered and training evaluations; and Future redeployment and training concerns*. Some of the redeployment implementation and training strategies documented in this review are: Skills-based redeployment, buddy support systems, and agreeing on locally specific principles, rather than strict procedures.

Conclusions: The COVID-19 pandemic presented unique challenges to develop flexible redeployment strategies and deliver training promptly while following infection control recommendations. This study synthesises original approaches to tackle these challenges, which are relevant to inform the development of targeted and adaptative training and redeployment plans considering the needs of staff.

Strengths and limitations of this study

- This is the first review to capture the redeployment and training strategies enlisted during the first year of the COVID-19 pandemic and the needs of redeployed staff.
- The study took the form of a systematic review which synthesised 20 papers identified across 9 databases.
- Papers included research studies (n=11), commentaries (n=5), and reports or guidelines (n=4), which were predominantly from the UK (n=8) and USA (n=5).
- Themes were identified through conducting a narrative synthesis on quantitative study outputs and thematic synthesis on qualitative study outputs, followed by combining the analyses through interpretive synthesis.
- Searches were limited to 9 databases and restricted from December 2019 to 8th of December 2020. The review will have missed literature published outside of these dates and not included in these databases.

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Introduction

To accommodate for the rapid fluctuations in the number of COVID-19 patients, healthcare organisations have been forced to optimise resource and staff allocation procedures. The unprecedented increase in the demand for intensive care services due to the COVID-19 pandemic involved the rapid redeployment of healthcare staff to these units. This posed multiple challenges, including devising new ways of working and rapid development and delivery of training. This review aimed to document key aspects of redeployment implementation, staff experiences and training.

Redeployment of staff from other specialties to intensive care aims to achieve the sustainable delivery of patient care by ensuring patients are cared for while maintaining staff wellbeing standards [1,2]. Redeployment can be used to facilitate the daily work of Intensive Care Units (ICU) when implementing task-based models, where key tasks of patient care (i.e. hygiene) are carried out promptly by competent redeployed staff [3]. Staff redeployment can also help to address staffing gaps caused by staff sickness and previous vacancies.

Building staff competence and confidence is an essential principle of safe redeployment [4]. Induction training must be provided to reacquaint redeployed staff with ICU ward multidisciplinary team practice, introduce them to clinical practices to care for COVID-19 patients and adequate use of Personal Protective Equipment (PPE) [5]. Time constraints and infection control measures pose unprecedented logistical challenges for training delivery, with traditional training methods such as those imparted in classrooms or at conferences not being possible. The impact of redeployment on staff wellbeing has also been identified. Redeployed staff have expressed concern in relation to their safety, the impact of their work on family members (including infecting them with the virus) and their own training and career progression [6,7].

A range of novel strategies to implement redeployment and share knowledge in the context of the pandemic have been proposed and trialled [8]. Gaining a detailed understanding of what worked and the needs that remain unaddressed will facilitate the development of redeployment plans.

Aim

The aim of this review was to provide a detailed understanding of the characteristics of redeployment to ICU and training provision during the first year of the COVID-19 pandemic. It sought to identify what worked in redeployment and training, and the concerns going forward with redeployment planning.

Research questions guiding the review were:

- What were the main strategies developed to redeploy staff to ICU?
- What were the principles of redeployment?
- What were redeployed staff experiences and perceived training needs?
- Were these needs addressed? If so, how? If not, why not?
- What were areas of good practice identified for both redeployment and training?

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Methods

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [9] and a protocol was developed a priori. The protocol was published on the authors' institutional website (see Supplementary Material 1) as it was not eligible for publication on PROSPERO.

Eligibility criteria

Studies and commentaries published in peer reviewed journals or official reports were included in this review if they were focused on training or redeployment to Intensive Care Units and related wards during COVID-19. Publication date was restricted from December 2019 to 8th of December 2020 (the date the search was conducted). There were no restrictions on language.

Articles were excluded if the focus was on redeployment to other areas of care, other viral infection emergencies, or changes in healthcare activities such as shifting to remote working.

Search strategy and study selection

Nine electronic databases were searched in December 2020 (including peer-reviewed and grey literature): Medline, CINAHL, PsychINFO, MedRxiv, Web of Science, The Health Management Consortium database, Social Science Research Network, OpenGrey and TRIP. The search strategy consisted of key terms referring to intensive care AND training AND redeployment AND healthcare workers. The search strategy was simplified when necessary for grey literature databases. A complete search strategy is provided in Supplementary Material 2.

Search results were imported into Rayyan [10] and deduplicated. Title and abstract screening was conducted independently by two researchers (CV and NVSJ) and discrepancies were resolved via discussion until consensus was reached. Full texts of articles deemed relevant for inclusion were then screened against full review eligibility criteria. The references of included full-text articles were reviewed to identify additional articles.

Data extraction and risk of bias assessment

NVSJ extracted the information of the articles onto a data extraction form developed on Qualtrics using a pre-defined list of data to extract (see Table 1). The sections of the list relevant to redeployment and training were created after a preliminary scan of the selected articles.

Table 1. List of data extracted from articles.

• Paper information: full citation; type of entry; location
• Study aims
• Population
• Study design: data collection and analysis methods

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- Redeployment: factors that determine redeployment; redeployment objectives; implementation strategies; redeployed staff experience; redeployment learnings; redeployed staff needs.
- Training: training programme offered; programme evaluation; pre/post results; what worked.

Study details such as setting, population and methodological characteristics were collected from all articles. The core findings collected from the studies included details of redeployment experiences and implementation strategies (research question 1), and training programmes offered (research question 2). Special attention was paid to extract information about lessons learned and concerns for the future.

We expected a heterogeneous group of studies using different questions and outcomes, therefore the AACODS (Authority, Accuracy, Coverage, Objectivity, Date, Significance) checklist [11] was applied during data extraction to assess the veracity of the source, clarity of the methods, acknowledgement of bias, and the relevance of the contribution to the field.

Data synthesis method

We conducted a narrative synthesis of the study characteristics and quantitative study outputs [12] and a framework-based thematic analysis of qualitative study outputs and grey literature [13,14]. Quantitative and qualitative results were combined using an interpretative synthesis to develop an understanding of how they related and answer the research questions [15].

Patient and public involvement

No patient was involved in the development of this study. The development of the research question and interpretation of the findings was informed by MC, JPJ and AM experience as clinicians and training providers during this pandemic. An infographic is being developed to facilitate the dissemination of findings outside of academic settings.

Results

Study selection

The screening and selection process is presented below in Figure 1 according to the PRISMA guidelines.

<Figure 1 about here>

Two articles were excluded at full text screening stage: Levy (2020), a commentary of one of the included articles which did not add information relevant to the research questions; and the study conducted by Chauhan (2020) which was not specific to redeployed staff.

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Study characteristics

From the twenty papers included in this systematic review, eleven (55%) were research studies; five (25%) were opinion pieces or commentaries; and four (20%) were reports or guidelines. Papers were primarily from the UK (N=8, 40%) and USA (N=5, 25%). Other locations included China, India, Germany and France. Eleven (55%) studies focused on redeployment implementation and experiences of redeployed staff, and nine (45%) on training delivery and evaluation, and dissemination of knowledge. The professional groups that were considered included nurses, anaesthetists, otolaryngologists, ophthalmologists, paramedics, orthopaedic surgeons, junior doctors, and physiotherapists.

From the eleven research studies included in the review, sample sizes ranged from ten [18] to 1269 [19]. One study reported 59% (N=19) of participants were male, and 41% (N=13) female [20]. Other participant demographic characteristics such as ethnicity were not provided.

All studies met 80% or more of the quality criteria assessed. This is, they were written by recognised experts, included reference lists, targeted a clear aim and stated details such as date, location and limitations.

A full summary of the studies included in this review can be found in Table 2.

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Table 2. Main characteristics of articles included in the review

First author name	Type of entry	Location	Article focus	Study design	Study population
NHS England and NHS Improvement (a)	Report/ guideline	UK	Management of surges: redeployment and training	N/A	N/A
Camilleri et al.	Scientific paper	UK and Ireland	Evaluation of remote learning course	Mixed methods: questionnaire design	Doctors, nurses, physiotherapists
George et al.	Scientific paper	USA	Redeployment and training of cardiac surgery practice	Review	Cardiac surgery divisions
Doussot et al.	Scientific paper	France	Redeployment feasibility and safety	Quantitative: prospective cohort study	Healthcare workers
Doyle et al.	Opinion piece/ commentary	England	Experiences of redeploying medical staff	N/A	N/A
Jansen et al.	Scientific paper	Germany	A blended learning concept	Mixed methods: questionnaire design	Medical personnel without ICU or ER training
Nair et al.	Report/ guideline	USA	Simulation-based training	N/A	N/A
Lim et al.	Scientific paper	UK	Redeployment of ophthalmologists	Quantitative: survey design	Ophthalmologists
Shipchandler et al.	Opinion piece/ commentary	USA	Role of otolaryngologists in redeployment	N/A	N/A
Fawcett et al.	Opinion piece/ commentary	UK	Education and scientific dissemination	N/A	N/A

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Marks et al.	Scientific paper	USA	Rapid deployment of critical care nurse education	Qualitative: interview design	Nurses
Kuang et al.	Scientific paper	China	Redeployment and training of non-specialists	Quantitative: cohort study	(a) non-specialists (b) specialists
Burnett et al.	Opinion piece/ commentary	USA	Deployment of an anesthesiology department	N/A	N/A
D'souza et al.	Report/ guideline	India	Adapting a secondary hospital for COVID-19	N/A	N/A
Brickman et al.	Scientific paper	USA	Rapid critical care training for nurses	Not specified	Nurses
Naik et al.	Scientific paper	USA	Telesimulation training for COVID-19 ventilator management	Not specified	Hospitalists, emergency medicine physicians and physician assistants, pediatric residents, nurses, and a nurse educator
Payne et al.	Scientific paper	UK	Redeployment of Surgical Trainees	Quantitative: questionnaire design	Surgical trainees
Coughlan et al.	Opinion piece/ commentary	UK	Redeployment of junior doctors	N/A	N/A
NHS England and NHS Improvement (b)	Report/ guideline	UK	Redeployment and safety	N/A	N/A
Hettle et al.	Scientific paper	UK	Cross-skill training and redeployment	Mixed methods:	Doctors

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questionnaire design

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Synthesis

Findings from this review are presented under the headings *Redeployment: Implementation strategies and learnings*; *Redeployed staff experience and strategies to address their needs*; *Redeployed staff learning needs*; *Training formats offered and training evaluations*.

Table 3 provides a summary of key findings.

Table 3. Summary of findings

Theme	Main findings
Redeployment: Implementation strategies and learnings	• Evolving landscape • Dependent on institute configuration, geographical location, population, and stage of the pandemic • Clear and consistent messaging through efficient channels critical in facilitating decision-making • Minimising of training needs, maximising use of redeployed staff's pre-existing skills • Redeployment into task-based groups with clear leadership to maximise efficiency and reduce risk
Redeployed staff experience and strategies to address their needs	• Staff anxiety and stress heighted by lack of support and working night shifts • Attention to support networks, availability of PPE, training, communication, protective legislation, and changing rota patterns. • Broader wellbeing needs: accommodation, access to testing, and family responsibilities • Opportunity to opt-out of redeployment or self-isolate without divulging personal information
Redeployed staff learning needs	• Enquiring about staff's previous experience and redeployment role • Training content to prioritise: PPE, infection control, ICU clinical assessments, mechanical ventilation, responding to acute respiratory distress syndrome, operating advance life support technology • Training needs specific to COVID-19 patient care: prone and positioning, maintaining vascular caterers and dialysis circuits, sedation and administering vasoactive medication • Liaising with family and communicating bad news
Training formats offered and training evaluations	• Blended approaches (theory + practice) • Online learning formats require follow-up practical sessions • Brief repeated training sessions rather than providing all information in one session • Training can be developed and delivered by shielding or recently retired staff • Training formats: online videos; live webinars; webcasts of conference presentations from previous years; podcasts and blogs to discuss recent publications; and infographics presenting main results • Training complimented by buddy systems which pair redeployed staff with experienced ICU staff • Iterative evaluation formats are most effective to develop training. Options include surveys, interviews, feedback sessions • Sharing learnings via scientific publications has significant limitations but innovation in peer review processes and dissemination proved useful

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Future redeployment and training concerns	• A need to return to practice as usual through de-escalation of redeployment • Practices with greater flexibility: the stepwise approach to redeployment; only redeploying discrete teams based on skills • Need for a balance between e-learning and face-to-face learning • Upcoming training focus shift to rehabilitation

Theme 1. Redeployment: Implementation strategies and learnings

Due to the shifting nature of the pandemic, areas of need were moving targets that constantly changed [21]. Redeployment varied depending on the individual institution configuration, geographical and population context and stage along the pandemic [2].

Clear decision-making processes were facilitated, for example, by clear definitions of "urgent" and efficient channelling of information to create a clear and consistent message [22]. For the latter, suggestions included limiting the use of email chains, using Dropbox or apps like Induction of Clinbee, and bespoke WhatsApp groups [2]. In the UK, Regional Emergency Preparedness Response and Resilience (EPRR) teams were reported as key to assisting hospitals in the management of acute surge. Clinicians and managers were advised to define warning triggers to anticipate a surge and establish communication lines with EPRR teams [2].

Redeployment planning aimed to minimise training needs and maximise use of redeployed staff's previous knowledge by placing staff in roles where their existing skills could be more easily transferrable [3]. A successful approach to redeployment was allocating redeployed staff to task-based groups, that is, multidisciplinary teams with clear leadership and constant communication, that aimed to complete a specific necessary step of intensive care when requested by experienced ICU staff [2,23]. This represented an important shift in ways of working and understanding collaborations between health specialists [3]. There were benefits in some specialists taking over ICU roles, examples of this were Otolaryngologists examining epistaxis, peritonsillar abscess, and facial trauma [21]; experienced renal physicians, together with trainee Radiologists developing line-insertion teams, or Orthopaedists and Physiotherapists assisting with proning [3]. Benefits included reduction of personnel required for procedures, reduction of aerosolization of the virus, shorter time dedicated to procedures, and minimal or no training required for them to provide assistance.

Theme 2. Redeployed staff experience and strategies to address their needs

Increasing staff buy-in was key for redeployment to work. Redeployed staff experienced anxiety and stress, particularly when lacking adequate support, during night shifts when less staff were available, and due to last minute rota changes. Staff wellbeing needs that required attention included accommodation, testing and family responsibilities [22,24]. Lim et al., (2020) reported that redeployed ophthalmologists' anxiety reduced once their redeployment role began. This was attributed to receiving support from staff in the redeployed area, the sufficient availability of PPE, and adequate training.

Coughlan et al. (2020) described solutions for stressors that Junior Doctors experienced due to working in unfamiliar ICU settings. The interpersonal communication required for intense multidisciplinary teamwork was facilitated by visual aids, anonymised

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whiteboards and the use of walkie-talkies. Worries about potential negligence proceedings resulting from working beyond their usual competencies were mitigated by emergency legislation to protect doctors.

Guidelines developed by NHS England & NHS Improvement (2020a) proposed addressing redeployed staff's needs by placing more experienced staff on nightshifts; encouraging questions; providing psychological support; accepting lower turnaround of patients; addressing issues about limited common areas and constant revision of rota patterns. Finally, all staff should be given the opportunity to opt-out of redeployment and self-isolate without divulging any personal information [27].

Theme 3. Redeployed staff learning needs

Redeployed staff learning needs varied depending on their previous experience and redeployment role. Training aimed to be targeted at the right level of difficulty depending on staff's most recent work experience and focused solely on content that will be relevant for redeployed roles. An example of prioritising learning objectives was teaching non-specialists to recognize worsening conditions and the need for ventilation, while specialists mastered details of the operation of ventilators [28].

Training needs considered essential to provide critical care services included learning the basics of ICU monitoring such as conducting and interpreting systematic clinical assessments; mechanical ventilation; response to acute respiratory distress syndrome (i.e., intubation, and cardiac arrest); and operating advanced life support technology.

Additionally, staff providing care specifically to COVID-19 patients required an introduction to diagnosis and anticipated patient needs. These included prone and positioning, maintaining vascular catheters and dialysis circuits, sedation, and administering vasoactive medication [4,19,23]. Redeployed staff were also often asked to liaise with families and required training on communicating bad news [20].

A particular emphasis was made on learning needs related to PPE and infection control, especially during aerosol-generating procedures [3,24,28].

Theme 4. Training formats offered and training evaluations

Most courses used blended approaches (theory + practice) and were collaboratively designed by a variety of clinical educators, intensive care experts and frontline staff [1,19,29,30]. Staff found it particularly useful when course content and practical sessions were repeated over time in consecutive sessions, rather than receiving an overwhelming amount of information in one session [18]. This required allocating time to access sustained training and senior staff pointing to most up to date training available [26].

Training development and delivery could be helped forward by recruiting as facilitators staff who were shielding or had recently retired [19]. Elder clinicians or those with other risk factors for severe COVID-19 infection could contribute to the pandemic response by leading simulation-based education sessions. Training formats included online videos; live webinars; webcasts of conference presentations from previous years; podcasts and blogs to discuss recent publications; and infographics presenting main results [8]. Online learning formats allowed staff to access material at their own pace

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and check understanding, however, there was a need for practical follow-up sessions to consolidate learning [1,18,19].

In addition, courses were generally complemented by buddy systems (pairing up redeployed staff with more experienced ICU staff) or other similar set-ups to provide support for redeployed staff during clinical practice [3,18]. Most training programmes were evaluated through surveys or interviews and one article reported the use of cycles of iteration composed of daily interactive feedback sessions with tutors and candidates to enable rapid improvement [19].

Sharing knowledge through scientific publications was thought to have significant limitations in the context of the pandemic. However, innovations in the peer review process and format of presenting results helped facilitate the continuation of this mode of sharing experiences [8]. Despite these efforts, reaching a single message for best practice was a significant challenge during the pandemic and affected training development. Guidelines recommended teaching principles rather than strict procedures to allow for adaptability as daily circumstances changed, as well as developing instructions at a local level [2].

Theme 5. Future redeployment and training concerns

As the pandemic evolved, the numbers of COVID-19 patients fluctuated rapidly and a need to return to 'business as usual' when possible became more apparent. For this reason, redeployment implementation strategies started to focus on facilitating de-escalating and escalating redeployment when necessary. Examples of practices that allowed for greater flexibility were the stepwise approach to redeployment, or only redeploying discrete teams based on skills (as described in the theme *Redeployment: Implementation strategies and learnings*) [3,27].

Regarding training, a need for balance between e-learning and face-to-face learning was identified. While e-learning is considered to have significant advantages such as a lower cost and reduced environmental impact, many feel there is still a need for face-to face learning. There are aspects of in-person meetings that enhance learning and wellbeing such as social interaction, hands-on teaching, and the opportunity to travel and visit venues, often incorporating some family downtime before or after the meeting [8].

Lastly, Camilleri et al. (2020) pointed out that as COVID-19 cases evolve, training focus will shift to rehabilitation.

Discussion

This review synthesised data from twenty studies to identify the core aspects of redeployment implementation, redeployed staff experiences and training during the first year of the COVID-19 pandemic.

Key principles for successful redeployment were developing staff work groups based on skills rather than specialty, maximising use of redeployed staff's transferable skills to minimise training, having a supportive work environment, and developing flexible arrangements that allowed for scaling redeployment up or down. How best to implement these principles depended on each individual institution's context, facilities, equipment, and the stage of the pandemic.

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Redeployed staff stressed the importance of counting with continuous support from more experienced staff. Inductions and sustained training were key and should be targeted to the right level depending on staff's previous experience. Central content included basics of ICU monitoring, response to acute respiratory distress syndrome, prone and positioning, and PPE donning and doffing. Staff assessed positively blended training with consecutive sessions.

To our knowledge this is the first review of literature about redeployment and training during this pandemic. Key strengths of this work include following the systematic approach of PRISMA guidelines and counting with frontline ICU clinicians as well as researchers among the authors. Searches were limited to 9 databases and restricted from December 2019 to 8th of December 2020. The review will have missed literature published outside of these dates and not included in these databases.

The challenges faced by redeployed staff during the COVID-19 pandemic overlapped with previous studies assessing experiences of junior staff in Emergency Departments, or staff redeployed to disaster and war zones (see **Table 4**) [31–33]. Theyyunni et al. (2013) analysed reflections from medical students after their emergency medicine rotation. The most common themes involved novice anxiety around critically ill patients and intubation procedures, miscommunication with other staff, and challenges stemming from the tension between textbook medicine and complex social situations. A challenge specific to the current pandemic was PPE use and everchanging guidelines [35]. PPE use resulted in difficulties with communication, responding to emergencies in a timely manner, and increased physical burden [36]. Other challenges were aspects outside of clinical practice that had an impact on redeployed staff's wellbeing, such as, access to breakout rooms and PCR testing, and family caring responsibilities [7,37].

Table 4. Key challenges for redeployment and training during the COVID-19 pandemic

- Efficient communication of clear and consistent messages in a rapidly changing environment.
- Challenges outside of clinical practice: family caring responsibilities, closure of leisure spaces.
- PPE availability and use.
- Delivering training in a timely manner while following infection control recommendations.
- Establishing flexible redeployment strategies to adapt to unprecedented variations in the number of patients requiring intensive care.

The essential role of training and need for innovative approaches has also been highlighted in the literature from previous virus outbreaks [38]. Multimodal, multidisciplinary and realistic simulation delivered in consecutive sessions are recommended options for successful training [29,39–41]. Doulias, Gallo, Rubio-Perez, Breukink, and Hahnloser (2020) summarised changes in surgical training during the COVID-19 pandemic. Key aspects of successful training overlapped with those identified for redeployment to ICU. The necessary shift to e-learning was possible due

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to innovative and collaborative approaches that mitigated the loss of access to other modes of learning during this time. In particular, interactive surgical simulation platforms offered a model of mentoring and continued guidance. Consultants were more available to impart simulation training and engage in discussions with trainees due to the reduced elective surgery services.

Recommendations for future redeployment plans include a shift towards flexible, innovative, and adaptive workforce management approaches. In the UK, the initial focus of the National Health Service (NHS) response to COVID-19 was establishing critical care capacity; this is now shifting towards developing pathways to support people to continue their rehabilitation and assessment in community settings [4,43]. This will require staff increasingly receiving training on regular remote monitoring, communication with patients and family, and remote end of life care.

Conclusion

Tackling ongoing challenges for healthcare provision in the current pandemic will require intense collaboration from multidisciplinary teams to build organisational resilience and optimise resources through successful execution of redeployment and training. Literature about healthcare provision in disaster contexts and warzones is the closest example of rapid redeployment to emergency care and can be a source of useful recommendations (such as the references cited in the discussion). However, the COVID-19 pandemic presented unique challenges and has resulted in original and innovative approaches which have been analysed in this review.

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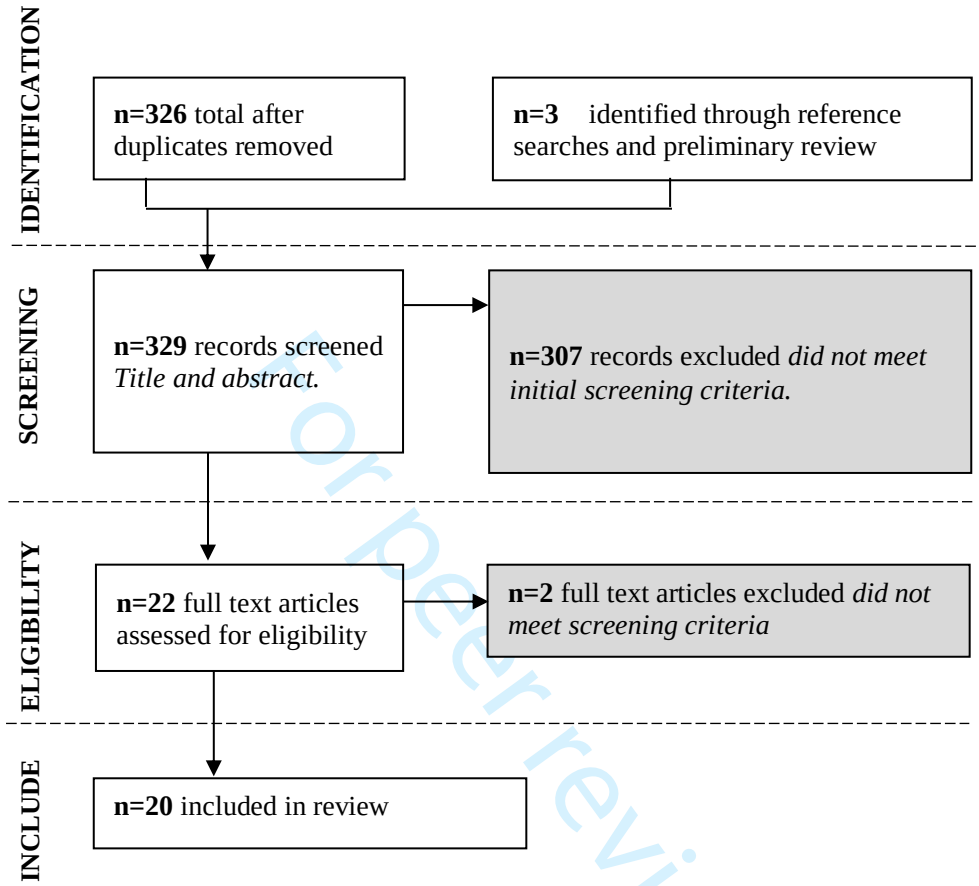
Figure Legends

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the screening and selection process conducted in this systematic review.

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Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the screening and selection process conducted in this systematic review.



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SUPPLEMENTARY MATERIALS

Supplementary Material 1: Study Protocol

Review title.

The redeployment of healthcare professionals to intensive care units (ICUs) during COVID-19 and the implementation of training support

3. * Anticipated or actual start date.

15 September, 2020

4. * Anticipated completion date.

31 June 2021

Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

1. Redeployment
    a. How was deployment/redeployment of staff defined during the COVID-19 pandemic?
    b. At what levels were staff redeployed? (hospital, local, regional, national)
    c. What were the main strategies developed to redeploy staff to ICUs?
2. Training
    a. What were the main training needs of redeployed staff?
    b. How did hospitals attempt to meet these needs?
    c. Were any training programmes developed at regional levels?
    d. Were any training programmes evaluated?

16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

Scientific databases:

- MEDLINE
- CINAHL Plus
- PsychInfo
- MedRxiv

Grey literature databases:

- Web of Science
- Social Science Research Network (SSRN)
- The Health Management Information Consortium (HMIC) database
- TRIP
- OpenGrey

Authors of relevant conference presentations identified in these searches will be contacted to enquire about possible unpublished studies.
Publication date will be restricted from December 2019 – present time.
There will be no language restrictions
MedRxiv search will be limited to the first 5 pages of output.

Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

Redeployment of healthcare professionals to intensive care and emergency services during COVID-19

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion

- Professionals redeployed to Intensive Care Units, A&E, Acute Medical Units and related wards during COVID-19.

Exclusion

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- Healthcare professionals redeployed to other areas of care.
- Redeployment during other viral infection emergencies.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion

- Redeployment during COVID-19

Exclusion

- Redeployment during other viral infection emergencies
- Other changes in healthcare professional activities (e.g. shift to remote working)

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

This review will not include a comparator/control group.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Inclusion

- Studies and commentaries published in peer reviewed journals: no restrictions based on study design (quantitative, qualitative, case studies and RCTs)
- Official reports and guidelines.

Exclusion

- Conference abstracts
- Grey literature such as personal blogs, commentaries from relatives/carers, or news articles
- Information on non-institutional or non-NGO websites

Context.

Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

This review will focus on Intensive Care Units, A&E, Acute Medical Units and related wards during COVID-19.

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurements are made, if these are part of the review inclusion criteria.

This review will provide a detailed understanding of the characteristics of redeployment of healthcare professionals during COVID-19. Potential outcomes will include:

- Principles and key objectives of deployment/redeployment of healthcare staff during COVID-19 (i.e. NHS guidelines for redeployment)
- Measures of staff experience and satisfaction with redeployment
- Results from pre/post tests in relation to training
- Evaluation of training programmes (i.e. Kirkpatrick model).

* Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not Applicable

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate
to the review

Not applicable

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

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Search results will be deduplicated and imported into Rayyan to manage the references for the tittle and abstract stages of the screening. Two researchers will review articles independently and discuss any discrepancies until consensus is reached.
At least two review authors will extract relevant information from included studies onto a data extraction form developed on REDCap using a pre-defined list of data to extract (see below). We will resolve any disagreement by discussion with a third member of the team.
The list of data to extract will be created after an initial review of the selected articles. We will extract descriptive data and any primary qualitative data or themes identified relating to redeployment and training for redeployed healthcare staff working in emergency services during COVID-19
Preliminary list of data to extract:

- Citation details
- Location of study
- Data collection methods / source of data (for secondary data analysis studies)
- Data analysis methods
- Population:
  - Professional group
  - Sample size
  - Sociodemographic and socioeconomic characteristics of the sample (gender, age, educational level, income level, and ethnicity)
- For research question 1: redeployment
  - Redeployment process
    - Objectives
    - Level at which it occurred (hospital, regional, national…)
    - Implementation strategies
    - Learnings – what worked
  - Redeployed staff
    - Training needs
    - Obligations
- For research question 2: Training
  - Training programmes offered/developed
  - Training programme evaluations
  - Learnings – what worked

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.
We expect a heterogeneous group of studies using different questions and outcomes, therefore a Mixed Methods Appraisal Tool to assess if the research question, methods, results and discussion align, alongside quality criteria specific to quantitative, qualitative and RCT studies will be applied. We will apply the AACODS checklist (Authority, Accuracy, Coverage, Objectivity, Date, Significance) to evaluate grey literature.

28. * Strategy for data synthesis.

*Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data. If meta- analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.*
We will synthesise qualitative study outputs applying framework-based thematic analysis (Braun & Clarke 2006; Gale et al. 2013), narrative synthesis of quantitative outputs (Popay et al. 2006), and an interpretative synthesis combining both results (Barnett-Page and Thomas, 2009).

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.
We do not anticipate it will be possible to conduct subgroup analysis. As part of our synthesis of results we will list differences between professional groups and locations.

Dissemination plans.

Do you intend to publish the review on completion?
Peer reviewed publication
Blog article

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Supplementary Material 2: Search strategy

Ovid: Medline, Psychinfo, Embase, and HMIC:

1: "Intensive care" OR "acute medical units" OR "critical care" OR "critically ill" OR "critical illness" OR "intensive treatment unit" OR "intensive therapy unit" OR "high dependency unit" or ICU or ITU or HDU
2. Training OR education OR course OR prepar* OR "clinical competence" OR "education intervention" OR "training intervention" OR Skill OR "competency-based education" OR "simulation" OR "simulation training"
3. Redeploy* OR deploy* OR reallocation OR reallocate
4. staff OR "Healthcare worker" OR HCW OR "healthcare professional" OR clinician OR nurse OR anaesthetist OR doctor OR trainee OR AHP OR allied health professional

Search 1 AND 2 AND 3 AND 4
Time limit Dec 2019-Dec 2020

MedRxiv
Redeploy* AND Covid-19

Web of Science

1: "Intensive care" OR "acute medical units" OR "critical care" OR "critically ill" OR "critical illness" OR "intensive treatment unit" OR "intensive therapy unit" OR "high dependency unit" or ICU or ITU or HDU
2. Training OR education OR course OR prepar* OR "clinical competence" OR "education intervention" OR "training intervention" OR Skill OR "competency-based education" OR "simulation" OR "simulation training"
3. Redeploy* OR deploy* OR reallocation OR reallocate
4. staff OR "Healthcare worker" OR HCW OR "healthcare professional" OR clinician OR nurse OR anaesthetist OR doctor OR trainee OR AHP OR allied health professional

Search 1 AND 2 AND 3 AND 4
Time limit to “last five years”

OpenGrey and SSRN

Redeploy* AND Covid-19
"coronavirus"

TRIP
"intensive care" AND training AND redeploy*

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5 (see supplementary material 1)
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5 (see supplementary material 2)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5-6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A

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PRISMA 2009 Checklist

Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6-7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14-16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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PRISMA 2009 Checklist

For more information, visit: **www.prisma-statement.org**.

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Training and Redeployment of healthcare professionals to Intensive Care during the COVID-19 pandemic: a systematic review

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Primary Subject Heading:	Intensive care
Secondary Subject Heading:	Medical education and training, Intensive care, Evidence based practice
Keywords:	COVID-19, HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Infection control < INFECTIOUS DISEASES, INTENSIVE & CRITICAL CARE, MEDICAL EDUCATION & TRAINING, Organisation of health services < HEALTH SERVICES ADMINISTRATION & MANAGEMENT

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Training and Redeployment of healthcare professionals to Intensive Care during the COVID-19 pandemic: a systematic review

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Abstract

Objectives: The rapid influx of COVID-19 patients to intensive care at a rate which exceeds pre-existing staff capacity has required the rapid development of innovative redeployment and training strategies, which not only considered patient care but also infection control. The aim of this study was to provide a detailed understanding of redeployment and training during the first year of the COVID-19 pandemic by capturing and considering the merit of the strategies enlisted and the experiences and needs of redeployed staff.

Design: The review involved a systematic search of key terms related to intensive care AND training AND redeployment AND healthcare workers within 9 databases (Medline, CINAHL, PsychINFO, MedRxiv, Web of Science, The Health Management Consortium database, Social Science Research Network, OpenGrey and TRIP) which took place on 16th July 2021. Analysis consisted of a narrative synthesis of quantitative study outputs, and a framework-based thematic analysis of qualitative study outputs and grey literature. These results were then combined applying an interpretative synthesis. We followed PRISMA and the review protocol was available online.

Results: Forty papers were analysed. These took place primarily in the UK (N=15, 37.5%) and USA (N=17, 42.5%). Themes presented in the results are *Redeployment: Implementation strategies and learnings; Redeployed staff experience and strategies to address their needs; Redeployed staff learning needs; Training formats offered and training evaluations; and Future redeployment and training delivery*. Based on this, key principles for successful redeployment and training are proposed.

Conclusions: The COVID-19 pandemic presents unique challenges to develop flexible redeployment strategies and deliver training promptly while following infection control recommendations. This study synthesises original approaches to tackle these challenges, which are relevant to inform the development of targeted and adaptative training and redeployment plans considering the needs of staff.

Strengths and limitations of this study

- This is the first review to capture the redeployment and training strategies enlisted during the first year of the COVID-19 pandemic and the needs of redeployed staff.
- The study took the form of a systematic review which synthesised 40 papers identified across 9 databases.
- Papers included research studies (N=21), commentaries (N=12), and reports or guidelines (N=7), which were predominantly from the UK (N=15) and USA (N=17).
- Themes were identified through conducting a narrative synthesis on quantitative study outputs and thematic synthesis on qualitative study outputs, followed by combining the analyses through interpretive synthesis.
- Searches were limited to 9 databases and restricted from December 2019 to 16 July 2021. The review will have missed literature published outside of these dates and not included in these databases.

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Introduction

To accommodate for the rapid fluctuations in the number of COVID-19 patients, healthcare organisations have been forced to optimise resource and staff allocation procedures. The unprecedented increase in the demand for intensive care services due to the COVID-19 pandemic involved the rapid redeployment of healthcare staff to these units. This posed multiple challenges, including devising new ways of working and rapid development and delivery of training. This review aimed to document key aspects of redeployment implementation, staff experiences and training.

Redeployment of staff from other specialties to intensive care aims to achieve the sustainable delivery of patient care by ensuring patients are cared for while maintaining staff wellbeing standards [1,2]. Redeployment can be used to facilitate the daily work of Intensive Care Units (ICU) when implementing task-based models, where key tasks of patient care (i.e. hygiene) are carried out promptly by competent redeployed staff [3]. Staff redeployment can also help to address staffing gaps caused by staff sickness and previous vacancies.

Building staff competence and confidence is an essential principle of safe redeployment [4]. Induction training must be provided to reacquaint redeployed staff with ICU ward multidisciplinary team practice, introduce them to clinical practices to care for COVID-19 patients and adequate use of Personal Protective Equipment (PPE) [5]. Time constraints and infection control measures pose unprecedented logistical challenges for training delivery, with traditional training methods such as those imparted in classrooms or at conferences not being possible. The impact of redeployment on staff wellbeing has also been identified [6,7]. Redeployed staff have expressed concern in relation to their safety, the impact of their work on family members (including infecting them with the virus) and their own training and career progression [6,8].

A range of novel strategies to implement redeployment and share knowledge in the context of the pandemic have been proposed and trialled [9]. Gaining a detailed understanding of what worked and the needs that remain unaddressed will facilitate the development of redeployment plans.

Aim

The aim of this review was to provide a detailed understanding of the characteristics of redeployment to ICU and training provision during the first year and a half of the COVID-19 pandemic. It sought to identify what worked in redeployment and training, and the concerns going forward with redeployment planning.

Research questions guiding the review were:

- What were the main strategies developed to redeploy staff to ICU?
- What were the principles of redeployment?
- What were redeployed staff experiences and perceived training needs?
- Were these needs addressed? If so, how? If not, why not?
- What were areas of good practice identified for both redeployment and training?

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Methods

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [10] and a protocol was developed a priori. The protocol was published on the authors' institutional website (see Supplementary Material 1) as it was not eligible for publication on PROSPERO.

Eligibility criteria

Studies and commentaries published in peer reviewed journals or official reports were included in this review if they were focused on training or redeployment to Intensive Care Units and related wards during COVID-19. Publication date was restricted from December 2019 to 16th of July 2021. There were no restrictions on language.

Articles were excluded if the focus was on redeployment to other areas of care, other viral infection emergencies, or changes in healthcare activities such as shifting to remote working.

Search strategy and study selection

Nine electronic databases were searched in December 2020 and again in July 2021 (including peer-reviewed and grey literature): Medline, CINAHL, PsychINFO, MedRxiv, Web of Science, The Health Management Consortium database, Social Science Research Network, OpenGrey and TRIP. The search strategy consisted of key terms referring to intensive care AND training AND redeployment AND healthcare workers. The search strategy was simplified when necessary for grey literature databases. A complete search strategy is provided in Supplementary Material 2.

Search results were imported into Rayyan [11] and deduplicated. Title and abstract screening was conducted independently by two researchers (CV and NVSJ for the first search, SEC and GS for the second search) and discrepancies were resolved via discussion until consensus was reached. Full texts of articles deemed relevant for inclusion were then screened against full review eligibility criteria. The references of included full-text articles were reviewed to identify additional articles. In the event that non-English papers were identified, members of the broader team who spoke the language were asked to review these.

Data extraction and risk of bias assessment

NVSJ and SEC extracted the information of the articles onto a data extraction form developed on Qualtrics using a pre-defined list of data to extract (see Table 1). The sections of the list relevant to redeployment and training were created after a preliminary scan of the selected articles.

Table 1. List of data extracted from articles.

• Paper information: full citation; type of entry; location
• Study aims
• Population

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- Study design: data collection and analysis methods
- Redeployment: factors that determine redeployment; redeployment objectives; implementation strategies; redeployed staff experience; redeployment learnings; redeployed staff needs.
- Training: training programme offered; programme evaluation; pre/post results; what worked.

Study details such as setting, population and methodological characteristics were collected from all articles. The core findings collected from the studies included details of redeployment experiences and implementation strategies (research question 1), and training programmes offered (research question 2). Special attention was paid to extract information about lessons learned and concerns for the future.

We expected a heterogeneous group of studies using different questions and outcomes, therefore the AACODS (Authority, Accuracy, Coverage, Objectivity, Date, Significance) checklist [12] was applied during data extraction to assess the veracity of the source, clarity of the methods, acknowledgement of bias, and the relevance of the contribution to the field.

Data synthesis method

We conducted a narrative synthesis of the study characteristics and quantitative study outputs [13] and a framework-based thematic analysis of qualitative study outputs and grey literature [14,15]. Quantitative and qualitative results were combined using an interpretative synthesis to develop an understanding of how they related and answer the research questions [16].

Patient and public involvement

No ICU patients were involved in the development of this study, however, all authors had first-hand experience of being infected, or having close relatives and friends who were infected with COVID-19. The development of the research question and interpretation of the findings was informed by MC, JPJ and AM experience as clinicians and training providers during this pandemic. An infographic is being developed to facilitate the dissemination of findings outside of academic settings.

Results

Study selection

The screening and selection process is presented below in Figure 1 according to the PRISMA guidelines.

<Figure 1 about here>

At full text screening stage forty-one articles were excluded as they did not add information relevant to the research questions; or they were not specific to redeployed staff; or they were abstract pieces from conferences.

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Study characteristics

From the forty papers included in this systematic review, twenty-one (52.5%) were research studies; twelve (30%) were opinion pieces or commentaries; and seven (17.5%) were reports or guidelines. Papers were primarily from the UK (N=15, 37.5%) and USA (N=17, 42.5%). Other locations included China, India, Belgium, Italy, Germany, France and two studies involving multiple countries. Fourteen (35%) studies focused on redeployment implementation and experiences of redeployed staff exclusively, and twenty-six (65%) on training delivery and evaluation, and dissemination of knowledge too. The professional groups that were considered included nurses, physician assistants, anaesthetists, otolaryngologists, ophthalmologists, paramedics, radiologists, urologists, neurologists, orthopaedic surgeons, pharmacy residents, junior doctors, physical therapists, occupational therapists, and physiotherapists.

From the twenty-one research studies included in the review, sample sizes ranged from ten [17] to 2527 [18]. One study reported 59% (N=19) of participants were male, and 41% (N=13) female [19], whilst another study reported the majority of the participants were female (70.8%, N=17) [20]. Other participant demographic characteristics such as ethnicity were not provided.

Of the studies thirty-five met 75% or more of the quality criteria assessed, this is, they were written by recognised experts, included reference lists, targeted a clear aim and stated details such as date, location and limitations. However, five of the studies only met 50-70% of the criteria as they did not have a clear aim and methodology, or they did not identify bias.

A full summary of the studies included in this review can be found in Supplementary Material 3.

Synthesis

Findings from this review are presented under the headings *Redeployment: Implementation strategies and learnings*; *Redeployed staff experience and strategies to address their needs*; *Redeployed staff learning needs*; *Training formats offered and training evaluations*, and distilled in a final section where we propose key principles for successful redeployment.

Table 3 provides a summary of key findings.

Table 3. Summary of findings

Theme	Main findings
Redeployment: Implementation strategies and learnings	• Evolving landscape • Dependent on institute configuration, geographical location, population, and stage of the pandemic • Clear and consistent messaging through efficient channels critical in facilitating decision-making • Minimising of training needs, maximising use of redeployed staff's pre-existing skills • Redeployment into task-based groups with clear leadership to maximise efficiency and reduce risk

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Redeployed staff experience and strategies to address their needs	• Staff anxiety and stress heighted by lack of support and working night shifts • Attention to support networks, availability of PPE, training, communication, protective legislation, and changing rota patterns • Broader wellbeing factors: accommodation, access to testing, and family responsibilities • Feelings of encouragement and changes in ways of thinking • Opportunity to opt-out of redeployment or self-isolate without divulging personal information
Redeployed staff learning needs	• Enquiring about staff's previous experience and redeployment role • Training content to prioritise: PPE, infection control, ICU clinical assessments, mechanical ventilation, responding to acute respiratory distress syndrome, operating advance life support technology • Training needs specific to COVID-19 patient care: prone and positioning, maintaining vascular caterers and dialysis circuits, sedation and administering vasoactive medication • Liaising with family and communicating bad news
Training formats offered and training evaluations	• Blended approaches (theory + practice) • Online learning formats require follow-up practical sessions • Brief repeated training sessions rather than providing all information in one session • Training can be developed and delivered by shielding or recently retired staff • Training formats: simulation sessions, online videos; live webinars; webcasts of conference presentations from previous years; competency checklists podcasts and blogs to discuss recent publications; and infographics presenting main results • Training complimented by buddy systems which pair redeployed staff with experienced ICU staff • Bedside learning co-ordinators shared changes in practice and checked understanding of healthcare staff • Iterative evaluation formats are most effective to develop training. Options include surveys, interviews, feedback sessions • Sharing learnings via scientific publications has significant limitations but innovation in peer review processes and dissemination proved useful
Future redeployment and training concerns	• A need to return to practice as usual through de-escalation of redeployment • Practices with greater flexibility: the stepwise approach to redeployment; only redeploying discrete teams based on skills • Need for a balance between e-learning and face-to-face learning • Upcoming training focus shift to rehabilitation

Theme 1. Redeployment: Implementation strategies and learnings

Due to the shifting nature of the pandemic, areas of need were moving targets that constantly changed [21]. Redeployment varied depending on the individual institution configuration, geographical and population context and stage along the pandemic [2]. Strategies to redistribute staff to meet surge demands varied across countries and health systems. In terms of sources from which to redeploy staff, for example, most places had access to volunteers for internal redeployment. Some of the places with the greatest burden of COVID-19 cases recruited from outside of the health system (e.g. traveling nurses) [22].

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Clear decision-making processes were facilitated, for example, by clear definitions of "urgent" and efficient channelling of information to create a clear and consistent message [23,24]. For the latter, suggestions included limiting the use of email chains, using online platforms such as Dropbox or apps like Induction of Clinbee, and bespoke WhatsApp groups [2,24–28]. In the UK, Regional Emergency Preparedness Response and Resilience (EPRR) teams were reported as key to assisting hospitals in the management of acute surge. Clinicians and managers were advised to define warning triggers to anticipate a surge and establish communication lines with EPRR teams [2]. Healthcare settings also made efforts to enable two-way communication to allow for feedback from redeployed staff through webinars, briefing meetings and feedback forms [24,25,27,29]. Transparency and involvement of frontline clinicians as stakeholders were important components and helped to maintain trust among staff [30].

Redeployment planning aimed to minimise training needs and maximise use of redeployed staff's previous knowledge by placing staff in roles where their existing skills could be more easily transferrable [3,24,29,31]. Panda et al. (2021) found that health system leaders (i.e. chief medical officers, chairs, or division chiefs) favoured a decentralised leadership approach, where organization-wide goals were disseminated to frontline leaders for implementation. This facilitated targeted training, as local leaders were better able to identify the training needs of each redeployee. A successful approach to redeployment was allocating redeployed staff to designated treatment teams; these are task-based groups, multidisciplinary teams with clear leadership and constant communication, that aimed to complete a specific necessary step of intensive care when requested by experienced ICU staff [2,24,26,27,30,32–36]. Similar to this is the tiered staffing model in which critical care-trained physicians or nurses oversaw non-ICU clinicians [30]. This represented an important shift in ways of working and understanding collaborations between health specialists [3]. There were benefits in some specialists taking over ICU roles, examples of this were Otolaryngologists examining epistaxis, peritonsillar abscess, and facial trauma [21]; experienced renal physicians, together with trainee Radiologists developing line-insertion teams, or Orthopaedists and Physiotherapists assisting with proning [3]. Benefits included reduction of personnel required for procedures, reduction of aerosolization of the virus, shorter time dedicated to procedures, and minimal or no training required for them to provide assistance.

A key barrier for successful redeployment planning was not being able to measure the need of human resources, this is, identifying which specific roles were in demand and which members of the workforce were available and healthy [22].

Theme 2. Redeployed staff experience and strategies to address their needs

Increasing staff buy-in was key for redeployment to work. Redeployed staff experienced anxiety and stress, particularly when lacking adequate support or PPE, during night shifts when less staff were available, and due to last minute rota changes [18,20,23,27–29,31,36–38]. Staff wellbeing needs that required attention included accommodation, food subsidisation, access to clean scrubs, parking, a 24-hour PPE hotline, testing and family responsibilities [23,24,29,37,39]. Lim et al., (2020) [40] reported that redeployed ophthalmologists' anxiety reduced once their redeployment role began. This was attributed to receiving support from staff in the redeployed area,

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the sufficient availability of PPE, and adequate training. Some staff often felt that the redeployment period had encouraged them to learn and change their ways of thinking in challenging situations [20,25].

Coughlan et al. (2020) [41] described solutions for stressors that Junior Doctors experienced due to working in unfamiliar ICU settings. The interpersonal communication required for intense multidisciplinary teamwork was facilitated by visual aids, anonymised whiteboards and the use of walkie-talkies. Worries about potential negligence proceedings resulting from working beyond their usual competencies were mitigated by emergency legislation to protect doctors.

Guidelines developed by NHS England & NHS Improvement (2020a) proposed addressing redeployed staff's needs by placing more experienced staff on nightshifts; encouraging questions; providing psychological support; accepting lower turnaround of patients; addressing issues about limited common areas and constant revision of rota patterns. Finally, all staff should be given the opportunity to opt-out of redeployment and self-isolate without divulging any personal information [42].

Theme 3. Redeployed staff learning needs

Redeployed staff learning needs varied depending on their previous experience and redeployment role. Training aimed to be targeted at the right level of difficulty depending on staff's most recent work experience and focused solely on content that was relevant for redeployed roles. An example of prioritising learning objectives was teaching non-specialists to recognize worsening conditions and the need for ventilation, while specialists mastered details of the operation of ventilators [43].

Across the literature, training needs considered essential to provide critical care services included learning the basics of ICU monitoring such as conducting and interpreting systematic clinical assessments; mechanical ventilation; response to acute respiratory distress syndrome (i.e., intubation, and cardiac arrest); conducting lung ultrasounds, management of circulatory shock, and operating advanced life support technology [e.g. 28,34,44].

Additionally, staff providing care specifically to COVID-19 patients required an introduction to diagnosis and anticipated patient needs. These included prone and positioning, maintaining vascular catheters and dialysis circuits, sedation, and administering vasoactive medication and continuous positive airway pressure services [4,27,32,33,36,44–46]. Redeployed staff were also often asked to liaise with families and required training on communicating bad news [19]. A particular emphasis was made on learning needs related to PPE and infection control, especially during aerosol-generating procedures [3,29,39,43,46]. Regarding the latter, studies reflected on the difficulty of ensuring staff safety due to permanently changing, and sometimes conflicting advice on the use of PPE [3,30]. As a result, teams assigned aerosol-generating procedures such as intubation were preferably staffed with experienced airway experts such as anaesthetists or otorhinolaryngologists to minimise the personnel required and the aerosolisation of the virus during procedures [21].

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Theme 4. Training formats offered and training evaluations

Most courses used blended approaches (theory + practice) and were collaboratively designed by a variety of clinical educators, intensive care experts and frontline staff [1,25,31,33–35,38,44,45,47,48]. Staff found it particularly useful when course content and practical sessions were repeated over time in consecutive sessions, rather than receiving an overwhelming amount of information in one session [17]. This required allocating time to access sustained training and senior staff pointing to most up to date training available [41].

Training development and delivery could be helped forward by recruiting as facilitators staff who were shielding or had recently retired [45]. Elder clinicians or those with other risk factors for severe COVID-19 infection could contribute to the pandemic response by leading simulation-based education sessions. Training formats included simulation sessions, online videos; live webinars; webcasts of conference presentations from previous years; competency checklists, podcasts and blogs to discuss recent publications; and infographics presenting main results [9,20,26,27,29,31,34,36,38,44]. Online learning formats allowed staff to access material at their own pace and check understanding, however, there was a need for practical follow-up sessions to consolidate learning [1,17,45].

In addition, courses were generally complemented by buddy systems (pairing up redeployed staff with more experienced ICU staff) or other similar set-ups to provide support for redeployed staff during clinical practice [3,17,31,34,36]. Most training programmes were evaluated through surveys or interviews and one article reported the use of cycles of iteration composed of daily interactive feedback sessions with tutors and candidates to enable rapid improvement [45]. Another article shared the role of bedside learning co-ordinators, who updated healthcare staff on any new changes to protocols and checked their understanding of these updates [25].

Sharing knowledge through scientific publications was thought to have significant limitations in the context of the pandemic. However, innovations in the peer review process and format of presenting results helped facilitate the continuation of this mode of sharing experiences [9]. Despite these efforts, reaching a single message for best practice was a significant challenge during the pandemic and affected training development. Guidelines recommended teaching principles rather than strict procedures to allow for adaptability as daily circumstances changed, as well as developing instructions at a local level [2].

Theme 5. Future redeployment and training delivery

As the pandemic evolved, the numbers of COVID-19 patients fluctuated rapidly and a need to return to ‘business as usual’ when possible was noted in the literature [4,23]. For this reason, redeployment implementation strategies started to focus on facilitating de-escalating and escalating redeployment when necessary. Examples of practices that allowed for greater flexibility were the stepwise approach to redeployment, or only redeploying discrete teams based on skills (as described in the theme *Redeployment: Implementation strategies and learnings*) [3,31,42].

Regarding training, a need for balance between e-learning and face-to-face learning was mentioned. While e-learning was considered to have significant advantages such as a

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lower cost and reduced environmental impact, many felt there is still a need for face-to face learning. There were aspects of in-person meetings that enhance learning and wellbeing such as social interaction, hands-on teaching, and the opportunity to travel and visit venues, often incorporating some family downtime before or after the meeting [9]. Lastly, Camilleri et al. (2020) pointed out that as COVID-19 cases evolve, training focus will shift to rehabilitation.

Proposed key principles for successful redeployment and training

Four key principles for successful redeployment emerged from the findings in this review: (1) developing staff work groups based on skills rather than specialty; (2) maximising use of redeployed staff's transferable skills to minimise training; (3) having a supportive work environment, including continuous support from more experienced staff; (4) and developing flexible arrangements that allowed for scaling redeployment up or down.

Inductions and sustained training were key. These should be targeted to the right level depending on staff's previous experience. Central training content included basics of ICU monitoring, response to acute respiratory distress syndrome, proning and positioning, and PPE donning and doffing. Staff assessed blended training (online + in-person practical sessions) taught in consecutive sessions in a positive way.

How best to implement these principles depended on each individual institution's context, facilities, equipment, and the stage of the pandemic.

Discussion

This review synthesised data from twenty studies to identify the core aspects of redeployment implementation, redeployed staff experiences and training during the first year and a half of the COVID-19 pandemic.

To our knowledge this is the first review of literature about redeployment and training during this pandemic. Key strengths of this work include following the systematic approach of PRISMA guidelines and counting with frontline ICU clinicians as well as researchers among the authors. Searches were conducted in a wide range of databases, including grey literature and preprint servers which have been an important source of information during the pandemic. However, searches were restricted from December 2019 to 16th of July 2021. The review will have missed literature published outside of these dates and not included in these databases. It is also possible that some studies from non-English speaking countries were not identified due to all the search terms being in English.

The challenges faced by redeployed staff during the COVID-19 pandemic overlapped with previous studies assessing experiences of junior staff in Emergency Departments, or staff redeployed to disaster and war zones (see **Table 4**) [49–51]. Theyyunni et al. (2013) [52] analysed reflections from medical students after their emergency medicine rotation. The most common themes involved novice anxiety around critically ill patients and intubation procedures, miscommunication with other staff, and challenges stemming from the tension between textbook medicine and complex social situations. Challenges specific to the current pandemic were PPE use and everchanging guidelines, and the paucity of testing at the initial stages [22,53]. PPE use resulted in difficulties

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with communication, responding to emergencies in a timely manner, and increased physical burden [54]. Also, frontline healthcare workers mentioned that not having testing available increased their anxiety about possibly infecting their families, and exacerbated their guilt when being off work due to suspected COVID-19 infections that could not be confirmed [6]. Other challenges included aspects outside of clinical practice that had an impact on redeployed staff's wellbeing, such as, access to breakout rooms, safety in lonely streets during stricter lockdowns, and family caring responsibilities [6,55]. Overall, effective healthcare leadership was found to consist not only of identifying and understanding the sources of fear and anxiety among individuals but also designing strategies to mitigate their effects [22].

Table 4. Key challenges for redeployment and training during the COVID-19 pandemic

- Efficient communication of clear and consistent messages in a rapidly changing environment.
- Challenges outside of clinical practice: family caring responsibilities, closure of leisure spaces.
- PPE availability and use.
- Delivering training in a timely manner while following infection control recommendations.
- Establishing flexible redeployment strategies to adapt to unprecedented variations in the number of patients requiring intensive care.

The essential role of training and need for innovative approaches has also been highlighted in the literature from previous virus outbreaks [56]. Multimodal, multidisciplinary and realistic simulation delivered in consecutive sessions are recommended options for successful training [48,57–59]. Doulias, Gallo, Rubio-Perez, Breukink, and Hahnloser (2020) [60] summarised changes in surgical training during the COVID-19 pandemic. Key aspects of successful training overlapped with those identified for redeployment to ICU. The necessary shift to e-learning was possible due to innovative and collaborative approaches that mitigated the loss of access to other modes of learning during this time. In particular, interactive surgical simulation platforms offered a model of mentoring and continued guidance. Consultants were more available to impart simulation training and engage in discussions with trainees due to the reduced elective surgery services.

Recommendations for future redeployment plans include a shift towards flexible, innovative, and adaptive workforce management approaches. In the UK, the initial focus of the National Health Service (NHS) response to COVID-19 was establishing critical care capacity; this is now shifting towards developing pathways to support people to continue their rehabilitation and assessment in community settings [4,61]. This will require staff increasingly receiving training on regular remote monitoring, communication with patients and family, and remote end of life care.

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Conclusion

Tackling ongoing challenges for healthcare provision in the current pandemic will require intense collaboration from multidisciplinary teams to build organisational resilience and optimise resources through successful execution of redeployment and training. Literature about healthcare provision in disaster contexts and warzones is the closest example of rapid redeployment to emergency care and can be a source of useful recommendations (such as the references cited in the discussion). However, the COVID-19 pandemic presents unique challenges and has resulted in original and innovative approaches which have been analysed in this review. Key principles of redeployment distilled from our findings can inform future practice for redeployment and training in ICU settings within the evolving context of COVID-19.

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Author contributions
NVSJ led the conception and design, acquisition of data, analysis, interpretation, drafting and revision of the manuscript.
CVP contributed to the conception and design, acquisition of data, analysis, interpretation, and drafting the manuscript.
GC and SEC contributed to the acquisition of data, analysis, interpretation, and drafting the manuscript.
MC, JPJ, AM contributed to the conception and design, interpretation, and drafting the manuscript.

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Data sharing agreement: All data relevant to the study are included in the article or uploaded as supplementary information. Further information is available upon reasonable request

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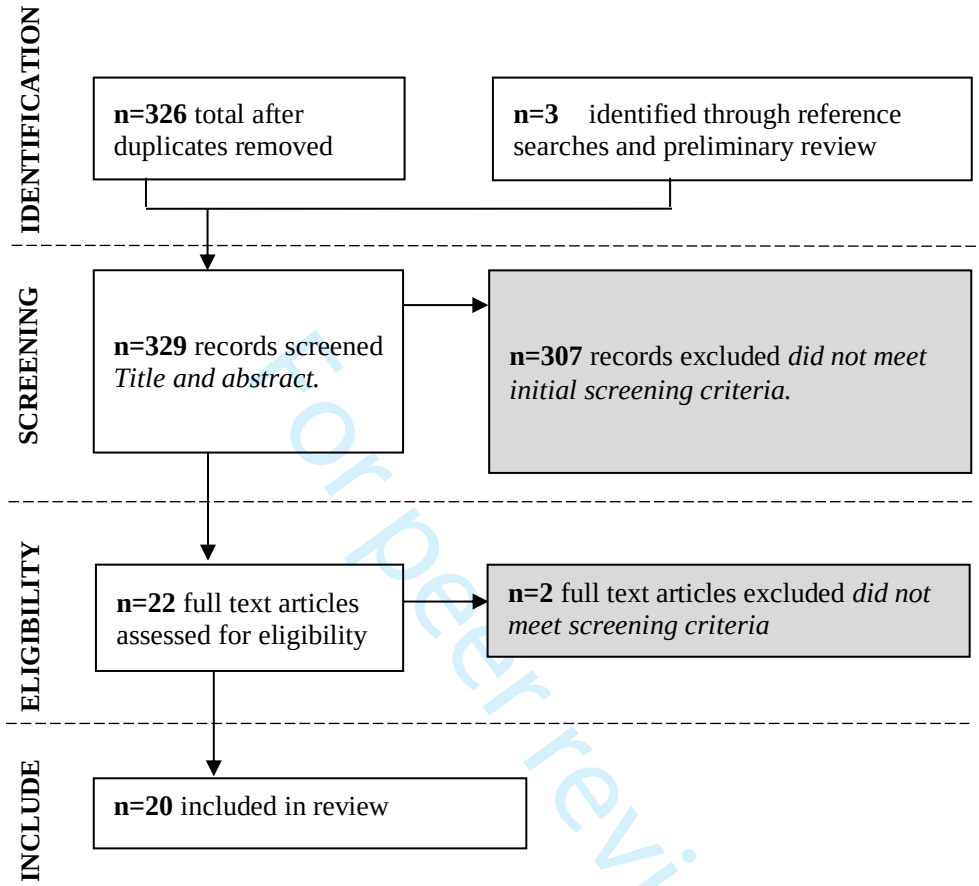
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Figure Legends

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the screening and selection process conducted in this systematic review.

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Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the screening and selection process conducted in this systematic review.



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SUPPLEMENTARY MATERIALS

Supplementary Material 1: Study Protocol

Review title.

The redeployment of healthcare professionals to intensive care units (ICUs) during COVID-19 and the implementation of training support

3. * Anticipated or actual start date.

15 September, 2020

4. * Anticipated completion date.

31 June 2021

Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

1. Redeployment
    a. How was deployment/redeployment of staff defined during the COVID-19 pandemic?
    b. At what levels were staff redeployed? (hospital, local, regional, national)
    c. What were the main strategies developed to redeploy staff to ICUs?
2. Training
    a. What were the main training needs of redeployed staff?
    b. How did hospitals attempt to meet these needs?
    c. Were any training programmes developed at regional levels?
    d. Were any training programmes evaluated?

16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

Scientific databases:

- MEDLINE
- CINAHL Plus
- PsychInfo
- MedRxiv

Grey literature databases:

- Web of Science
- Social Science Research Network (SSRN)
- The Health Management Information Consortium (HMIC) database
- TRIP
- OpenGrey

Authors of relevant conference presentations identified in these searches will be contacted to enquire about possible unpublished studies.

Publication date will be restricted from December 2019 – present time.

There will be no language restrictions

MedRxiv search will be limited to the first 5 pages of output.

Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

Redeployment of healthcare professionals to intensive care and emergency services during COVID-19

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion

- Professionals redeployed to Intensive Care Units, A&E, Acute Medical Units and related wards during COVID-19.

Exclusion

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- Healthcare professionals redeployed to other areas of care.
- Redeployment during other viral infection emergencies.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion

- Redeployment during COVID-19

Exclusion

- Redeployment during other viral infection emergencies
- Other changes in healthcare professional activities (e.g. shift to remote working)

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

This review will not include a comparator/control group.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Inclusion

- Studies and commentaries published in peer reviewed journals: no restrictions based on study design (quantitative, qualitative, case studies and RCTs)
- Official reports and guidelines.

Exclusion

- Conference abstracts
- Grey literature such as personal blogs, commentaries from relatives/carers, or news articles
- Information on non-institutional or non-NGO websites

Context.

Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.

This review will focus on Intensive Care Units, A&E, Acute Medical Units and related wards during COVID-19.

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurements are made, if these are part of the review inclusion criteria.

This review will provide a detailed understanding of the characteristics of redeployment of healthcare professionals during COVID-19. Potential outcomes will include:

- Principles and key objectives of deployment/redeployment of healthcare staff during COVID-19 (i.e. NHS guidelines for redeployment)
- Measures of staff experience and satisfaction with redeployment
- Results from pre/post tests in relation to training
- Evaluation of training programmes (i.e. Kirkpatrick model).

* Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.

Not Applicable

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate
to the review

Not applicable

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.

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Search results will be deduplicated and imported into Rayyan to manage the references for the tittle and abstract stages of the screening. Two researchers will review articles independently and discuss any discrepancies until consensus is reached.
At least two review authors will extract relevant information from included studies onto a data extraction form developed on REDCap using a pre-defined list of data to extract (see below). We will resolve any disagreement by discussion with a third member of the team.
The list of data to extract will be created after an initial review of the selected articles. We will extract descriptive data and any primary qualitative data or themes identified relating to redeployment and training for redeployed healthcare staff working in emergency services during COVID-19
Preliminary list of data to extract:

- Citation details
- Location of study
- Data collection methods / source of data (for secondary data analysis studies)
- Data analysis methods
- Population:
  - Professional group
  - Sample size
  - Sociodemographic and socioeconomic characteristics of the sample (gender, age, educational level, income level, and ethnicity)
- For research question 1: redeployment
  - Redeployment process
    - Objectives
    - Level at which it occurred (hospital, regional, national…)
    - Implementation strategies
    - Learnings – what worked
  - Redeployed staff
    - Training needs
    - Obligations
- For research question 2: Training
  - Training programmes offered/developed
  - Training programme evaluations
  - Learnings – what worked

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.
We expect a heterogeneous group of studies using different questions and outcomes, therefore a Mixed Methods Appraisal Tool to assess if the research question, methods, results and discussion align, alongside quality criteria specific to quantitative, qualitative and RCT studies will be applied. We will apply the AACODS checklist (Authority, Accuracy, Coverage, Objectivity, Date, Significance) to evaluate grey literature.

28. * Strategy for data synthesis.

*Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data. If meta- analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.*
We will synthesise qualitative study outputs applying framework-based thematic analysis (Braun & Clarke 2006; Gale et al. 2013), narrative synthesis of quantitative outputs (Popay et al. 2006), and an interpretative synthesis combining both results (Barnett-Page and Thomas, 2009).

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.
We do not anticipate it will be possible to conduct subgroup analysis. As part of our synthesis of results we will list differences between professional groups and locations.

Dissemination plans.

Do you intend to publish the review on completion?
Peer reviewed publication
Blog article

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Supplementary Material 2: Search strategy

Ovid: Medline, Psychinfo, Embase, and HMIC:

1: "Intensive care" OR "acute medical units" OR "critical care" OR "critically ill" OR "critical illness" OR "intensive treatment unit" OR "intensive therapy unit" OR "high dependency unit" or ICU or ITU or HDU
2. Training OR education OR course OR prepar* OR "clinical competence" OR "education intervention" OR "training intervention" OR Skill OR "competency-based education" OR "simulation" OR "simulation training"
3. Redeploy* OR deploy* OR reallocation OR reallocate
4. staff OR "Healthcare worker" OR HCW OR "healthcare professional" OR clinician OR nurse OR anaesthetist OR doctor OR trainee OR AHP OR allied health professional

Search 1 AND 2 AND 3 AND 4
Time limit Dec 2019-Dec 2020
Time limit Dec 2020-July 2021

MedRxiv
Redeploy* AND Covid-19

Web of Science

1: "Intensive care" OR "acute medical units" OR "critical care" OR "critically ill" OR "critical illness" OR "intensive treatment unit" OR "intensive therapy unit" OR "high dependency unit" or ICU or ITU or HDU
2. Training OR education OR course OR prepar* OR "clinical competence" OR "education intervention" OR "training intervention" OR Skill OR "competency-based education" OR "simulation" OR "simulation training"
3. Redeploy* OR deploy* OR reallocation OR reallocate
4. staff OR "Healthcare worker" OR HCW OR "healthcare professional" OR clinician OR nurse OR anaesthetist OR doctor OR trainee OR AHP OR allied health professional

Search 1 AND 2 AND 3 AND 4
Time limit to “last five years”

OpenGrey and SSRN

Redeploy* AND Covid-19
"coronavirus"

TRIP
"intensive care" AND training AND redeploy*

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Supplementary Material 3

Table 2. Main characteristics of articles included in the review

First author name	Type of entry	Location	Article focus	Study design	Study population
NHS England and NHS Improvement (a)	Report/ guideline	UK	Management of surges: redeployment and training	N/A	N/A
Camilleri et al.	Scientific paper	UK and Ireland	Evaluation of remote learning course	Mixed methods: questionnaire design	Doctors, nurses, physiotherapists
George et al.	Scientific paper	USA	Redeployment and training of cardiac surgery practice	Review	Cardiac surgery divisions
Doussot et al.	Scientific paper	France	Redeployment feasibility and safety	Quantitative: prospective cohort study	Healthcare workers
Doyle et al.	Opinion piece/ commentary	England	Experiences of redeploying medical staff	N/A	N/A
Jansen et al.	Scientific paper	Germany	A blended learning concept	Mixed methods: questionnaire design	Medical personnel without ICU or ER training
Nair et al.	Report/ guideline	USA	Simulation-based training	N/A	N/A
Lim et al.	Scientific paper	UK	Redeployment of ophthalmologists	Quantitative: survey design	Ophthalmologists
Shipchandler et al.	Opinion piece/ commentary	USA	Role of otolaryngologists in redeployment	N/A	N/A

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Fawcett et al.	Opinion piece/ commentary	UK	Education and scientific dissemination	N/A	N/A
Marks et al.	Scientific paper	USA	Rapid deployment of critical care nurse education	Qualitative: interview design	Nurses
Kuang et al.	Scientific paper	China	Redeployment and training of non-specialists	Quantitative: cohort study	(a) non-specialists (b) specialists
Burnett et al.	Opinion piece/ commentary	USA	Deployment of an anesthesiology department	N/A	N/A
D'souza et al.	Report/ guideline	India	Adapting a secondary hospital for COVID-19	N/A	N/A
Brickman et al.	Scientific paper	USA	Rapid critical care training for nurses	Not specified	Nurses
Naik et al.	Scientific paper	USA	Telesimulation training for COVID-19 ventilator management	Not specified	Hospitalists, emergency medicine physicians and physician assistants, pediatric residents, nurses, and a nurse educator
Payne et al.	Scientific paper	UK	Redeployment of surgical trainees	Quantitative: questionnaire design	Surgical trainees
Coughlan et al.	Opinion piece/ commentary	UK	Redeployment of junior doctors	N/A	N/A

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NHS England and NHS Improvement (b)	Report/ guideline	UK	Redeployment and safety	N/A	N/A
Hettle et al.	Scientific paper	UK	Cross-skill training and redeployment	Mixed methods: questionnaire design	Doctors
Bohmer et al.	Opinion piece/ commentary	UK	Learning system to manage workforce and training of redeployed staff	N/A	Redeployed staff with ICU experience and volunteers with and without clinical experience
Khusid et al.	Scientific paper	USA	Redeployment of urology residents	Quantitative: questionnaire design	Urology residents
Shi et al.	Scientific paper	USA	Redeployment and training of radiology trainees	Quantitative: retrospective interventional study	Radiology trainees
KCE Belgian Health Care Knowledge Centre	Report/ guideline	Belgium	Redeployment and training strategies throughout Belgian hospitals	N/A	Medical staff
NHS GIRFT	Report/ guideline	UK	Redeployment and training recommendations for NHS hospitals	N/A	NHS medical staff
Panda et al.	Scientific paper	USA, UK, New Zealand, Singapore & South Korea	Recounts from hospital leaders in 5 countries on the redeployment strategies that were used	Qualitative: interview design	Hospital leaders
Raith et al.	Report/ guideline	UK	Redeployment and training of neurology staff and ward nurses	N/A	Neurology staff and nurses with and

					without ICU experience
Leng et al.	Opinion piece/ commentary	UK	Redeployment and training of staff from the burns, hands, and plastics department	N/A	Doctors
Uchida et al.	Opinion piece/ commentary	USA	Redeployment and training of pharmacy residents	N/A	Pharmacy residents
Danielis et al.	Scientific paper	Italy	Redeployment of nurses	Qualitative: descriptive study	Registered nurses
Yuriditsky et al.	Scientific paper	USA	Redeployment and simulation training of medical staff	Quantitative: survey design	Cardiologists, surgeons, hospitalists, nurse practitioners, physician associates and chief medical residents
Siva et al.	Opinion piece/ commentary	UK	Redeployment of Pediatric ICU trainees and training on adult intensive care	N/A	PICU trainees
Ch'ang et al.	Scientific paper	USA	Video based learning curriculum to teach critical care concepts to neurology trainees	Interventional study with quantitative survey design	Neuroscience trainees
Hickey et al.	Opinion piece/ commentary	USA	Redeployment of registered nurses and emergency department nurses	N/A	Registered nurses and emergency department nurses
DiMaggio et al.	Scientific paper	USA	Impact of redeployment on mental health of staff	Quantitative: survey design	Physicians, nurse practitioners and physician associates

Robinson et al.	Opinion piece/ commentary	UK	Redeployment and training of respiratory and infectious disease healthcare staff to a new ward-based team	N/A	Respiratory and infectious disease healthcare staff
Wells et al.	Opinion piece/ commentary	USA	Redeployment and training of nurses, technologists, and existing ICU staff	N/A	Registered nurses and technologists
Khajuria et al.	Scientific paper	41 countries	Impact of redeployment and training on mental health of staff	Cross-sectional study with quantitative survey design	Doctors, nurses, allied healthcare professionals
Chiu et al.	Scientific paper	USA	Review of the methods and training used to develop the Prone Team using redeployed staff	N/A	Physical and occupational therapists and registered nurses
Vranas et al.	Scientific paper	USA	Recounts from intensivists on the redeployment and training strategies used to redeploy non-ICU staff	Qualitative: interview design	Anesthesiologists and surgeons and other non-ICU clinicians

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5 (see supplementary material 1)
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5 (see supplementary material 2)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5-6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A

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Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6-7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14-16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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Training and Redeployment of Healthcare Workers to Intensive Care Units (ICUs) During the COVID-19 Pandemic: a Systematic Review

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Training and Redeployment of healthcare workers to Intensive Care Units (ICUs) during the COVID-19 pandemic: a systematic review

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Abstract

Objectives: The rapid influx of COVID-19 patients to intensive care at a rate which exceeds pre-existing staff capacity has required the rapid development of innovative redeployment and training strategies, which not only considered patient care but also infection control. The aim of this study was to provide a detailed understanding of redeployment and training during the first year of the COVID-19 pandemic by capturing and considering the merit of the strategies enlisted and the experiences and needs of redeployed healthcare workers (HCWs).

Design: The review involved a systematic search of key terms related to intensive care AND training AND redeployment AND healthcare workers within 9 databases (Medline, CINAHL, PsychINFO, MedRxiv, Web of Science, The Health Management Consortium database, Social Science Research Network, OpenGrey and TRIP) which took place on 16th July 2021. Analysis consisted of a narrative synthesis of quantitative study outputs, and a framework-based thematic analysis of qualitative study outputs and grey literature. These results were then combined applying an interpretative synthesis. We followed PRISMA and the review protocol was available online.

Results: Forty papers were analysed. These took place primarily in the UK (N=15, 37.5%) and USA (N=17, 42.5%). Themes presented in the results are *Redeployment: Implementation strategies and learnings; Redeployed HCWs experience and strategies to address their needs; Redeployed HCWs learning needs; Training formats offered and training evaluations; and Future redeployment and training delivery*. Based on this, key principles for successful redeployment and training are proposed.

Conclusions: The COVID-19 pandemic presents unique challenges to develop flexible redeployment strategies and deliver training promptly while following infection control recommendations. This study synthesises original approaches to tackle these challenges, which are relevant to inform the development of targeted and adaptative training and redeployment plans considering the needs of HCWs.

Strengths and limitations of this study

- This is the first review to capture the redeployment and training strategies enlisted during the first year of the COVID-19 pandemic and the needs of redeployed HCWs.
- The study took the form of a systematic review which synthesised 40 papers identified across 9 databases.
- Papers included research studies (N=21), commentaries (N=12), and reports or guidelines (N=7), which were predominantly from the UK (N=15) and USA (N=17) (limiting the exploration of redeployment in Low- and Middle-Income Countries).
- Themes were identified through conducting a narrative synthesis on quantitative study outputs and thematic synthesis on qualitative study outputs, followed by combining the analyses through interpretive synthesis.

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- Searches were limited to 9 databases and restricted from December 2019 to 16 July 2021. The review will have missed literature published outside of these dates and not included in these databases.

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Introduction

To accommodate for the rapid fluctuations in the number of COVID-19 patients, healthcare organisations have been forced to optimise resource and staff allocation procedures. The unprecedented increase in the demand for intensive care services due to the COVID-19 pandemic involved the rapid redeployment of healthcare HCWs to these units. This posed multiple challenges, including devising new ways of working and rapid development and delivery of training. This review aimed to document key aspects of redeployment implementation, HCW experiences and training.

Redeployment of HCWs from other specialties to intensive care can be used to achieve the sustainable delivery of patient care [1,2]. Redeployment can be used to facilitate the daily work of Intensive Care Units (ICU) when implementing task-based models, where key tasks of patient care (i.e. hygiene) are carried out promptly by competent redeployed HCWs [3]. Staff redeployment can also help to address staffing gaps caused by staff sickness and previous vacancies.

Building HCWs' competence and confidence is an essential principle of safe redeployment [4]. Induction training must be provided to reacquaint redeployed HCWs with ICU ward multidisciplinary team practice, introduce them to clinical practices to care for COVID-19 patients and adequate use of Personal Protective Equipment (PPE) [5]. Time constraints and infection control measures pose unprecedented logistical challenges for training delivery, with traditional training methods such as those imparted in classrooms or at conferences not being possible. The impact of redeployment on HCW wellbeing has also been identified [6,7]. Redeployed HCWs have expressed concern in relation to their safety, the impact of their work on family members (including infecting them with the virus) and their own training and career progression [6,8].

A range of novel strategies to implement redeployment and share knowledge in the context of the pandemic have been proposed and trialled [9]. Gaining a detailed understanding of what worked and the needs that remain unaddressed will facilitate the development of redeployment plans for this pandemic as well as future ones.

Aim

The aim of this review was to provide a detailed understanding of the characteristics of redeployment to ICU and training provision during the first year and a half of the COVID-19 pandemic. It sought to identify what worked in redeployment and training, and the concerns going forward with redeployment planning.

Research questions guiding the review were:

- What were the main strategies developed to redeploy HCWs to ICU?
- What were the principles of redeployment?
- What were redeployed HCWs' experiences and perceived training needs?
- Were these needs addressed? If so, how? If not, why not?
- What were areas of good practice identified for both redeployment and training?

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Methods

This review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement [10] and a protocol was developed a priori. The protocol was published on the authors' institutional website (see Supplementary Material 1) as it was not eligible for publication on PROSPERO.

Eligibility criteria

Studies and commentaries published in peer reviewed journals or official reports were included in this review if they focused on training or redeployment to ICUs and related wards during COVID-19. The publication dates were restricted from December 2019 to 16th of July 2021. There were no restrictions on language.

Articles were excluded if the focus was on redeployment to other areas of care, other viral infection emergencies, or changes in healthcare activities such as shifting to remote working.

Search strategy and study selection

Nine electronic databases were searched in December 2020 and again in July 2021 (including peer-reviewed and grey literature): Medline, CINAHL, PsychINFO, MedRxiv, Web of Science, The Health Management Consortium database, Social Science Research Network, OpenGrey and TRIP. The search strategy consisted of key terms referring to intensive care AND training AND redeployment AND healthcare workers. The search strategy was simplified when necessary for grey literature databases. A complete search strategy is provided in Supplementary Material 2.

Search results were imported into Rayyan [11] and deduplicated. Title and abstract screening was conducted independently by two researchers (CV and NVSJ for the first search, SEC and GC for the second search) and discrepancies were resolved via discussion until consensus was reached. Full texts of articles deemed relevant for inclusion were then screened against full review eligibility criteria. The references of included full-text articles were reviewed to identify additional articles. In the event that non-English papers were identified, members of the broader team who spoke the language were asked to review these.

Data extraction and risk of bias assessment

NVSJ and SEC extracted the information of the articles using a data extraction form developed on Qualtrics based on a pre-defined list of data (see Table 1). The sections of the list relevant to redeployment and training were created after a preliminary scan of the selected articles.

Table 1. List of data extracted from articles.

• Paper information: full citation; type of entry; location
• Study aims
• Population

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- Study design: data collection and analysis methods
- Redeployment: factors that determine redeployment; redeployment objectives; implementation strategies; redeployed HCWs' experience; redeployment learnings; redeployed HCWs' needs.
- Training: training programme offered; programme evaluation; pre/post results; what worked.

Study details such as setting, population and methodological characteristics were collected from all articles. The core findings collected from the studies included details of redeployment experiences and implementation strategies (research question 1), and training programmes offered (research question 2). Special attention was paid to extract information about lessons learned and concerns for the future.

We expected a heterogeneous group of studies using different questions and outcomes, therefore the AACODS (Authority, Accuracy, Coverage, Objectivity, Date, Significance) checklist [12] was applied during data extraction to assess the veracity of the source, clarity of the methods, acknowledgement of bias, and the relevance of the contribution to the field.

Data synthesis method

We conducted a narrative synthesis of the study characteristics and quantitative study outputs [13] and a framework-based thematic analysis of qualitative study outputs and grey literature [14,15]. Quantitative and qualitative results were combined using an interpretative synthesis to develop an understanding of how they related and answer the research questions [16,17]. This is, identifying emerging patterns, or lines of argument, across studies by repeatedly comparing them, and integrating these lines of argument interpretatively to provide a bigger picture [17].

Patient and public involvement

No ICU patients were involved in the development of this study, however, all authors had first-hand experience of being infected, or having close relatives and friends who were infected with COVID-19. The development of the research question and interpretation of the findings was informed by MC, JPJ and AM, who worked as clinicians and training providers during this pandemic. An infographic was developed to facilitate the dissemination of findings outside of academic settings.

Results

Study selection

The screening and selection process is presented below in Figure 1 according to the PRISMA guidelines.

<Figure 1 about here>

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At full text screening stage, forty-one articles were excluded as these did not include information relevant to the research questions; or were not specific to redeployed HCWs; or were abstract pieces from conferences.

Study characteristics

From the forty papers included in this systematic review, twenty-one (52.5%) were research studies; twelve (30%) were opinion pieces or commentaries; and seven (17.5%) were reports or guidelines. Papers were primarily from the UK (N=15, 37.5%) and USA (N=17, 42.5%). Other locations included China, India, Belgium, Italy, Germany, France and two studies involving multiple countries. Fourteen (35%) studies focused on redeployment implementation and experiences of redeployed HCWs exclusively, and twenty-six (65%) on training delivery and evaluation, and dissemination of knowledge. The professional groups that were considered included nurses, physician assistants, anaesthetists, otolaryngologists, ophthalmologists, paramedics, radiologists, urologists, neurologists, orthopaedic surgeons, pharmacy residents, junior doctors, physical therapists, occupational therapists, and physiotherapists.

From the twenty-one research studies included in the review, sample sizes ranged from ten [18] to 2527 [19]. One study reported 59% (N=19) of participants were male, and 41% (N=13) female [20], whilst another study reported the majority of the participants were female (70.8%, N=17) [21]. Other participant demographic characteristics such as ethnicity were not provided.

Of the studies, thirty-five met 75% or more of the quality criteria assessed as these were written by recognised experts, included reference lists, targeted a clear aim and stated details such as date, location and limitations. However, five of the studies only met 50-70% of the criteria as they did not have a clear aim and methodology, or did not identify bias.

A full summary of the studies included in this review can be found in Table 2.

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Table 2.

First author name	Type of paper	Location	Article focus	Study design	Study population
NHS England and NHS Improvement (a)	Report/ guideline	UK	Management of surges: redeployment and training	N/A	N/A
Camilleri et al.	Scientific paper	UK and Ireland	Evaluation of remote learning course	Mixed methods: questionnaire design	Doctors, nurses, physiotherapists
George et al.	Scientific paper	USA	Redeployment and training of cardiac surgery practice	Review	Cardiac surgery divisions
Doussot et al.	Scientific paper	France	Redeployment feasibility and safety	Quantitative: prospective cohort study	Healthcare workers
Doyle et al.	Opinion piece/ commentary	England	Experiences of redeploying medical staff	N/A	N/A
Jansen et al.	Scientific paper	Germany	A blended learning concept	Mixed methods: questionnaire design	Medical personnel without ICU or ER training
Nair et al.	Report/ guideline	USA	Simulation-based training	N/A	N/A
Lim et al.	Scientific paper	UK	Redeployment of ophthalmologists	Quantitative: survey design	Ophthalmologists
Shipchandler et al.	Opinion piece/ commentary	USA	Role of otolaryngologists in redeployment	N/A	N/A
Fawcett et al.	Opinion piece/ commentary	UK	Education and scientific dissemination	N/A	N/A

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Marks et al.	Scientific paper	USA	Rapid deployment of critical care nurse education	Qualitative: interview design	Nurses
Kuang et al.	Scientific paper	China	Redeployment and training of non-specialists	Quantitative: cohort study	(a) non-specialists (b) specialists
Burnett et al.	Opinion piece/ commentary	USA	Deployment of an anesthesiology department	N/A	N/A
D'souza et al.	Report/ guideline	India	Adapting a secondary hospital for COVID-19	N/A	N/A
Brickman et al.	Scientific paper	USA	Rapid critical care training for nurses	Not specified	Nurses
Naik et al.	Scientific paper	USA	Telesimulation training for COVID-19 ventilator management	Not specified	Hospitalists, emergency medicine physicians and physician assistants, pediatric residents, nurses, and a nurse educator
Payne et al.	Scientific paper	UK	Redeployment of surgical trainees	Quantitative: questionnaire design	Surgical trainees
Coughlan et al.	Opinion piece/ commentary	UK	Redeployment of junior doctors	N/A	N/A
NHS England and NHS Improvement (b)	Report/ guideline	UK	Redeployment and safety	N/A	N/A
Hettle et al.	Scientific paper	UK	Cross-skill training and redeployment	Mixed methods:	Doctors

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				questionnaire design	
Bohmer et al.	Opinion piece/ commentary	UK	Learning system to manage workforce and training of redeployed staff	N/A	Redeployed staff with ICU experience and volunteers with and without clinical experience
Khusid et al.	Scientific paper	USA	Redeployment of urology residents	Quantitative: questionnaire design	Urology residents
Shi et al.	Scientific paper	USA	Redeployment and training of radiology trainees	Quantitative: retrospective interventional study	Radiology trainees
KCE Belgian Health Care Knowledge Centre	Report/ guideline	Belgium	Redeployment and training strategies throughout Belgian hospitals	N/A	Medical staff
NHS GIRFT	Report/ guideline	UK	Redeployment and training recommendations for NHS hospitals	N/A	NHS medical staff
Panda et al.	Scientific paper	USA, UK, New Zealand, Singapore & South Korea	Recounts from hospital leaders in 5 countries on the redeployment strategies that were used	Qualitative: interview design	Hospital leaders
Raith et al.	Report/ guideline	UK	Redeployment and training of neurology staff and ward nurses	N/A	Neurology staff and nurses with and without ICU experience
Leng et al.	Opinion piece/ commentary	UK	Redeployment and training of staff from the burns, hands, and plastics department	N/A	Doctors
Uchida et al.	Opinion piece/ commentary	USA	Redeployment and training of pharmacy residents	N/A	Pharmacy residents

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Danielis et al.	Scientific paper	Italy	Redeployment of nurses	Qualitative: descriptive study	Registered nurses
Yuriditsky et al.	Scientific paper	USA	Redeployment and simulation training of medical staff	Quantitative: survey design	Cardiologists, surgeons, hospitalists, nurse practitioners, physician associates and chief medical residents
Siva et al.	Opinion piece/ commentary	UK	Redeployment of Pediatric ICU trainees and training on adult intensive care	N/A	PICU trainees
Ch'ang et al.	Scientific paper	USA	Video based learning curriculum to teach critical care concepts to neurology trainees	Interventional study with quantitative survey design	Neuroscience trainees
Hickey et al.	Opinion piece/ commentary	USA	Redeployment of registered nurses and emergency department nurses	N/A	Registered nurses and emergency department nurses
DiMaggio et al.	Scientific paper	USA	Impact of redeployment on mental health of staff	Quantitative: survey design	Physicians, nurse practitioners and physician associates
Robinson et al.	Opinion piece/ commentary	UK	Redeployment and training of respiratory and infectious disease healthcare staff to a new ward-based team	N/A	Respiratory and infectious disease healthcare staff
Wells et al.	Opinion piece/ commentary	USA	Redeployment and training of nurses, technologists, and existing ICU staff	N/A	Registered nurses and technologists
Khajuria et al.	Scientific paper	41 countries	Impact of redeployment and training on mental health of staff	Cross-sectional study with quantitative survey design	Doctors, nurses, allied healthcare professionals

Chiu et al.	Scientific paper	USA	Review of the methods and training used to develop the Prone Team using redeployed staff	N/A	Physical and occupational therapists and registered nurses
Vranas et al.	Scientific paper	USA	Recounts from intensivists on the redeployment and training strategies used to redeploy non-ICU staff	Qualitative: interview design	Anesthesiologists and surgeons and other non-ICU clinicians

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Synthesis

Findings from this review are presented under the headings *Redeployment: Implementation strategies and learnings*; *Redeployed HCWs' experience and strategies to address their needs*; *Redeployed HCWs' learning needs*; *Training formats offered and training evaluations* and distilled in a final section where we propose key principles for successful redeployment.

Table 3 provides a summary of the themes and their description.

Table 3. Summary of findings

Theme	Description of the theme	References
Redeployment: Implementation strategies	• Redeployment strategies depended on institute configuration, geographical location, population, and stage of the pandemic • Clear and consistent messaging through efficient channels was considered critical in facilitating decision-making • A useful strategy involved maximising the use of redeployed HCWs' pre-existing skills • Many papers reported redeployment of HCWs into task-based groups with clear leadership to maximise efficiency and reduce risk	[2, 21-24, 25-36]
Redeployed staff experience and strategies to address their needs	• HCWs' anxiety and stress were heighted by lack of support and working night shifts • Redeployed HCWs' main needs were good support networks, availability of PPE, training, and good communication • Factors that affected HCWs' wellbeing were: accommodation, access to testing, and family responsibilities • Redeployment strategies could involve giving staff the opportunity to opt-out of redeployment or self-isolate without divulging personal information	[18,20,23,-25,,27-31,36-42]
Redeployed staff learning needs	• Training should be tailored to HCWs' previous experience and redeployment role • Training programmes can prioritise the following content: PPE, infection control, ICU clinical assessments, mechanical ventilation, responding to acute respiratory distress syndrome, operating advance life support technology, liaising with families and communicating bad news	[3,4,19,21,27-30,32-34,36,39,43-46]

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	• Staff training needs specific to COVID-19 patient care include: proning and positioning, maintaining vascular caterers and dialysis circuits, sedation and administering vasoactive medication	
Training formats offered and training evaluations	• HCWs preferred training with blended approaches (theory + practice) • If training is based solely on online learning formats, it requires follow-up practical sessions • HCWs preferred repeated shorter training sessions rather than receiving all information in one session • Training can be developed and delivered by shielding or recently retired HCWs • Training formats reported in the literature included: simulation sessions, online videos; live webinars; webcasts of conference presentations from previous years; competency checklists podcasts and blogs to discuss recent publications; and infographics presenting main results • Helpful training strategies included: buddy systems which paired redeployed HCWs with experienced ICU HCWs, bedside learning co-ordinators shared changes in practice and checked understanding with HCWs • Iterative evaluation formats are most effective to develop training. Options included surveys, interviews, feedback sessions	[1,3,9,17,20,25-27,29,31,33-36,38,44,45,47,48]
Future redeployment and training concerns	• Redeployment for future patient surges can be based on a stepwise approach to redeployment; only redeploying discrete teams based on skills • Training needs to continue after redeployment to maintain skills (using blended approaches)Future training is likely to focus on patient rehabilitation	[3,4,9,23,31,42]

Theme 1. Redeployment: Implementation strategies

Due to the shifting nature of the pandemic, areas of need were moving targets that constantly changed [22]. Redeployment varied depending on the individual institution configuration, geographical and population context and stage along the pandemic [2].

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Strategies to redistribute HCWs to meet surge demands varied across countries and health systems. In terms of sources from which to redeploy HCWs, for example, most places had access to volunteers for internal redeployment. Some of the places with the greatest burden of COVID-19 cases recruited from outside of the health system (e.g. traveling nurses) [23].

Clear decision-making processes were facilitated, for example, by clear definitions of "urgent" and efficient channelling of information to create a clear and consistent message [24,25]. For the latter, suggestions included limiting the use of email chains, using online platforms such as Dropbox or apps like Induction of Clinbee, and bespoke WhatsApp groups [2,25–29]. In the UK, Regional Emergency Preparedness Response and Resilience (EPRR) teams were reported as key to assisting hospitals in the management of acute surges. Clinicians and managers were advised to define warning triggers to anticipate a surge and establish communication lines with EPRR teams [2]. Healthcare settings also made efforts to enable two-way communication to allow for feedback from redeployed HCWs through webinars, briefing meetings and feedback forms [25,26,28,30]. Transparency and involvement of frontline clinicians as stakeholders were important components and helped to maintain trust among HCWs [31].

Redeployment planning aimed to minimise training needs and maximise the use of redeployed HCWs' previous knowledge by placing HCWs in roles where their existing skills could be more easily transferrable [3,25,30,32]. Panda et al. (2021) found that health system leaders (i.e. chief medical officers, chairs, or division chiefs) favoured a decentralised leadership approach, where organization-wide goals were disseminated to frontline leaders for implementation. This facilitated targeted training, as local leaders were better able to identify the training needs of each redeployee. A successful approach to redeployment was allocating redeployed HCWs to designated treatment teams; these are task-based groups, multidisciplinary teams with clear leadership and constant communication, that aimed to complete a specific necessary step of intensive care when requested by experienced ICU HCWs [2,25,27,28,31,33–37]. Another successful strategy was a tiered staffing model where critical care-trained physicians or nurses oversaw non-ICU clinicians [31]. This represented an important shift in ways of working and understanding collaborations between health specialists [3]. There were benefits in some specialists taking over ICU roles, examples of this were Otolaryngologists examining epistaxis, peritonsillar abscess, and facial trauma [22]; experienced renal physicians, together with trainee Radiologists developing line-insertion teams, or Orthopaedists and Physiotherapists assisting with proning [3]. Benefits included the reduction of personnel required for procedures, reduction of aerosolization of the virus, shorter time dedicated to procedures, and minimal or no training required for them to provide assistance.

A key barrier for successful redeployment planning was not being able to measure the need of human resources, this is, identifying which specific roles were in demand and which members of the workforce were available and healthy [23].

Theme 2. Redeployed HCW experience and strategies to address their needs

Increasing staff buy-in was key for redeployment to work. Redeployed HCWs experienced anxiety and stress, particularly when lacking adequate support or PPE,

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during night shifts when fewer HCWs were available, and due to last minute rota changes [19,21,24,28–30,32,37–39]. HCW wellbeing needs that required attention included accommodation, food subsidisation, access to clean scrubs, parking, a 24-hour PPE hotline, testing and family responsibilities [24,25,30,38,40]. Lim et al., (2020) [41] reported that redeployed ophthalmologists' anxiety reduced once their redeployment role began. This was attributed to receiving support from HCWs in the redeployed area, the sufficient availability of PPE, and adequate training. Some HCWs often felt that the redeployment period had encouraged them to learn and change their ways of thinking in challenging situations [21,26].

Coughlan et al. (2020) [42] described solutions for stressors that Junior Doctors experienced due to working in unfamiliar ICU settings. The interpersonal communication required for intense multidisciplinary teamwork was facilitated by visual aids, anonymised whiteboards and the use of walkie-talkies. Worries about potential negligence proceedings resulting from working beyond their usual competencies were mitigated by emergency legislation to protect doctors.

Guidelines developed by NHS England & NHS Improvement (2020a) proposed addressing redeployed HCWs' needs by placing more experienced HCWs on nightshifts; encouraging questions; providing psychological support; accepting lower turnaround of patients; addressing issues about limited common areas and constant revision of rota patterns. One paper argued that HCWs should be given the opportunity to opt-out of redeployment and self-isolate without divulging any personal information [43].

Theme 3. Redeployed HCWs' learning needs

Redeployed HCWs' learning needs varied depending on their previous experience and redeployment role. Training needed to be targeted at the right level of difficulty depending on HCWs' most recent work experience and focused solely on content that was relevant for redeployed roles. An example of prioritising learning objectives was teaching non-specialists to recognize worsening conditions and the need for ventilation, while specialists mastered details of the operation of ventilators [44].

Across the literature, training needs considered essential to provide critical care services included learning the basics of ICU monitoring such as conducting and interpreting systematic clinical assessments; mechanical ventilation; response to acute respiratory distress syndrome (i.e., intubation, and cardiac arrest); conducting lung ultrasounds, management of circulatory shock, and operating advanced life support technology [e.g. 28,34,44].

Additionally, HCWs providing care specifically to COVID-19 patients required an introduction to diagnosis and anticipated patient needs. These included prone and positioning, maintaining vascular catheters and dialysis circuits, sedation, and administering vasoactive medication and continuous positive airway pressure services [4,28,33,34,37,45–47]. Redeployed HCWs were also often asked to liaise with families and required training on communicating bad news [20]. A particular emphasis was made on learning needs related to PPE and infection control, especially during aerosol-generating procedures [3,30,40,44,47]. Regarding the latter, studies reflected on the difficulty of ensuring HCW safety due to permanently changing, and sometimes

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conflicting advice on the use of PPE [3,31]. As a result, teams assigned aerosol-generating procedures such as intubation were preferably staffed with experienced airway experts such as anaesthetists or otorhinolaryngologists to minimise the personnel required and the aerosolisation of the virus during procedures [22].

Theme 4. Training formats offered and training evaluations

Most courses reported in the literature included blended approaches (theory + practice) and were collaboratively designed by a variety of clinical educators, intensive care experts and frontline HCWs [1,26,32,34–36,39,45,46,48,49]. HCWs found it particularly useful when course content and practical sessions were repeated over time in consecutive sessions, rather than receiving an overwhelming amount of information in one session [18]. This required allocating time to access training on a continuous basis and making sure it was kept up-to-date [42].

Some of the training development and delivery was carried out by HCWs who were shielding or had recently retired [46]. Elder clinicians or those with other risk factors for severe COVID-19 infection contributed to the pandemic response by leading simulation-based education sessions. Training formats included simulation sessions, online videos; live webinars; webcasts of conference presentations from previous years; competency checklists, podcasts and blogs to discuss recent publications; and infographics presenting main results [9,21,27,28,30,32,35,37,39,45]. Online learning formats allowed HCWs to access material at their own pace and check understanding, however, there was a need for practical follow-up sessions to consolidate learning [1,18,46].

In addition, courses were generally complemented by buddy systems (pairing up redeployed HCWs with more experienced ICU HCWs) or other similar set-ups to provide support for redeployed HCWs during clinical practice [3,18,32,35,37]. Most training programmes were evaluated through surveys or interviews and one article reported the use of cycles of iteration composed of daily interactive feedback sessions with tutors and candidates to enable rapid improvement [46]. Another article shared the role of bedside learning co-ordinators, who updated HCWs on any new changes to protocols and checked their understanding of these updates [26].

Innovations in the peer review process and format of presenting results helped to disseminate scientific evidence in a timely way [9]. However, reaching a consistent message for best practice was a significant challenge during the pandemic and affected training development. Guidelines recommended teaching principles rather than strict procedures to allow for adaptability to the local context and as circumstances changed.[2].

Theme 5. Future redeployment and training delivery

As the pandemic evolved, the numbers of COVID-19 patients fluctuated rapidly and a need to return to ‘business as usual’ when possible was noted in the literature [4,24]. For this reason, redeployment implementation strategies started to focus on facilitating de-escalating and escalating redeployment when necessary. Examples of practices that allowed for greater flexibility were the stepwise approach to redeployment, or only redeploying discrete teams based on skills (as described in the theme *Redeployment: Implementation strategies and learnings*) [3,32,43].

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Regarding training, a need for balance between e-learning and face-to-face learning was mentioned. While e-learning was considered to have significant advantages such as a lower cost and reduced environmental impact, many felt there was still a need for face-to face learning. There were aspects of in-person meetings that enhanced learning and wellbeing such as social interaction, hands-on teaching, and the opportunity to travel and visit venues [9]. Lastly, Camilleri et al. (2020) pointed out that as COVID-19 cases evolve, training focus will shift to rehabilitation.

Proposed key principles for successful redeployment and training

Four key principles for successful redeployment emerged from the findings in this review: (1) developing HCW work groups based on skills rather than specialty; (2) maximising the use of redeployed HCWs' transferable skills to minimise training; (3) having a supportive work environment, including continuous support from more experienced HCWs; (4) and developing flexible arrangements that allowed for scaling redeployment up or down.

Inductions and sustained training were key. These should be targeted to the right level depending on HCWs' previous experience. Central training content included the basics of ICU monitoring, response to acute respiratory distress syndrome, proning and positioning, and PPE donning and doffing. HCWs assessed blended training (online + in-person practical sessions) taught in consecutive sessions in a positive way.
The implementation of these principles depended on each individual institution's context, facilities, equipment, and the stage of the pandemic.

Discussion

This review synthesised data from forty studies to identify the core aspects of redeployment implementation, redeployed HCWs' experiences and training during the first year and a half of the COVID-19 pandemic.

To our knowledge, this is the first review of literature about redeployment and training during this pandemic. Key strengths of this work include following the systematic approach of PRISMA guidelines and combining authors with systematic review as well as clinical expertise. Searches were conducted in a wide range of databases, including grey literature and preprint servers, which have been important sources of information during the pandemic. However, searches were restricted from December 2019 to 16th of July 2021. The review will have missed literature published outside of these dates and not included in these databases. It is possible that some studies from non-English speaking countries were not identified due to all the search terms being in English. It is also important to mention that most of the included papers were from High-Income Countries, limiting our exploration of the experiences of Low and Middle Income Countries (LMICs).

The challenges faced by redeployed HCWs during the COVID-19 pandemic overlapped with previous studies assessing experiences of junior HCWs in Emergency Departments, or HCWs redeployed to disaster and war zones [50–52]. Theyyunni et al. (2013) [53] analysed reflections from medical students after their emergency medicine rotation. The most common themes involved novice anxiety around critically ill patients and intubation procedures, miscommunication with other HCWs, and challenges

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stemming from the tension between textbook medicine and complex social situations. Challenges specific to the current pandemic were PPE use and everchanging guidelines, and the paucity of testing at the initial stages [23,54]. PPE use resulted in difficulties with communication, responding to emergencies in a timely manner, and increased physical burden [55]. Also, frontline HCWs mentioned that not having testing available increased their anxiety about possibly infecting their families, and exacerbated their guilt when being off work due to suspected COVID-19 infections that could not be confirmed [6]. Other challenges included aspects outside of clinical practice that had an impact on redeployed HCWs' wellbeing, such as, access to breakout rooms, safety in lonely streets during stricter lockdowns, and family caring responsibilities [6,56]. Overall, effective healthcare leadership was found to consist not only of identifying and understanding the sources of fear and anxiety among individuals but also designing strategies to mitigate their effects [23].

The essential role of training and need for innovative approaches has also been highlighted in the literature from previous virus outbreaks [57]. Multimodal, multidisciplinary and realistic simulation delivered in consecutive sessions were recommended options for successful training [49,58–60]. Doulias, Gallo, Rubio-Perez, Breukink, and Hahnloser (2020) [61] summarised changes in surgical training during the COVID-19 pandemic. Key aspects of successful training overlapped with those identified for redeployment to ICU. The necessary shift to e-learning was possible due to innovative and collaborative approaches that mitigated the loss of access to other modes of learning during this time. In particular, interactive surgical simulation platforms offered a model of mentoring and continued guidance. Consultants were more available to impart simulation training and engage in discussions with trainees due to the reduced elective surgery services.

Recommendations for future redeployment plans included a shift towards flexible, innovative, and adaptive workforce management approaches. In the UK, the initial focus of the National Health Service (NHS) response to COVID-19 was establishing critical care capacity; this is now shifting towards developing pathways to support people to continue their rehabilitation and assessment in community settings [4,62]. This will require that HCWs receive training on regular remote monitoring, communication with patients and family, and remote end of life care.

Conclusion

Tackling ongoing challenges for healthcare provision in the current pandemic will require intense collaboration from multidisciplinary teams to build organisational resilience and optimise resources through successful execution of redeployment and training. Literature about healthcare provision in disaster contexts and warzones is the closest example of rapid redeployment to emergency care and can be a source of useful recommendations. However, the COVID-19 pandemic presents unique challenges and has resulted in original and innovative approaches, which have been analysed in this review.

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The key principles of redeployment distilled from our findings can inform future practice for redeployment and training in ICU settings within the evolving context of COVID-19 and future pandemics. Furthermore, similar strategies could be implemented to mitigate the negative effects of the workforce shortage crisis that affects many countries around the world.

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Author contributions
NVSJ led the conception and design, acquisition of data, analysis, interpretation, drafting and revision of the manuscript.
CVP contributed to the conception and design, acquisition of data, analysis, interpretation, and drafting the manuscript.
GC and SEC contributed to the acquisition of data, analysis, interpretation, and drafting the manuscript.
MC, JPJ, AM contributed to the conception and design, interpretation, and drafting the manuscript.

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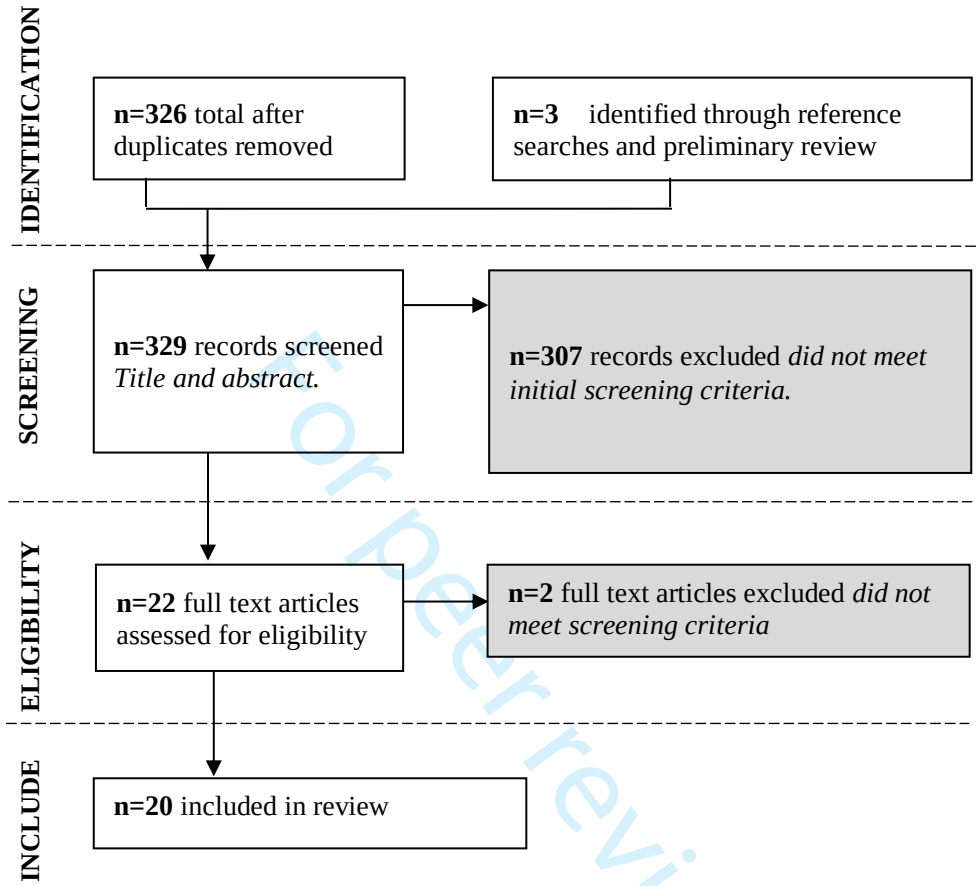
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Figure Legends

Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the screening and selection process conducted in this systematic review.

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Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram of the screening and selection process conducted in this systematic review.



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SUPPLEMENTARY MATERIALS

Supplementary Material 1: Study Protocol

Review title.

The redeployment of healthcare professionals to intensive care units (ICUs) during COVID-19 and the implementation of training support

3. * Anticipated or actual start date.

15 September, 2020

4. * Anticipated completion date.

31 June 2021

Review question.

State the review question(s) clearly and precisely. It may be appropriate to break very broad questions down into a series of related more specific questions. Questions may be framed or refined using PI(E)COS or similar where relevant.

1. Redeployment
    a. How was deployment/redeployment of staff defined during the COVID-19 pandemic?
    b. At what levels were staff redeployed? (hospital, local, regional, national)
    c. What were the main strategies developed to redeploy staff to ICUs?
2. Training
    a. What were the main training needs of redeployed staff?
    b. How did hospitals attempt to meet these needs?
    c. Were any training programmes developed at regional levels?
    d. Were any training programmes evaluated?

16. * Searches.

State the sources that will be searched (e.g. Medline). Give the search dates, and any restrictions (e.g. language or publication date). Do NOT enter the full search strategy (it may be provided as a link or attachment below.)

Scientific databases:

- MEDLINE
- CINAHL Plus
- PsychInfo
- MedRxiv

Grey literature databases:

- Web of Science
- Social Science Research Network (SSRN)
- The Health Management Information Consortium (HMIC) database
- TRIP
- OpenGrey

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Authors of relevant conference presentations identified in these searches will be contacted to enquire about possible unpublished studies.
Publication date will be restricted from December 2019 – present time.
There will be no language restrictions
MedRxiv search will be limited to the first 5 pages of output.

Condition or domain being studied.

Give a short description of the disease, condition or healthcare domain being studied in your systematic review.

Redeployment of healthcare professionals to intensive care and emergency services during COVID-19

19. * Participants/population.

Specify the participants or populations being studied in the review. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion

- Professionals redeployed to Intensive Care Units, A&E, Acute Medical Units and related wards during COVID-19.

Exclusion

- Healthcare professionals redeployed to other areas of care.
- Redeployment during other viral infection emergencies.

20. * Intervention(s), exposure(s).

Give full and clear descriptions or definitions of the interventions or the exposures to be reviewed. The preferred format includes details of both inclusion and exclusion criteria.

Inclusion

- Redeployment during COVID-19

Exclusion

- Redeployment during other viral infection emergencies
- Other changes in healthcare professional activities (e.g. shift to remote working)

21. * Comparator(s)/control.

Where relevant, give details of the alternatives against which the intervention/exposure will be compared (e.g. another intervention or a non-exposed control group). The preferred format includes details of both inclusion and exclusion criteria.

This review will not include a comparator/control group.

22. * Types of study to be included.

Give details of the study designs (e.g. RCT) that are eligible for inclusion in the review. The preferred format includes both inclusion and exclusion criteria. If there are no restrictions on the types of study, this should be stated.

Inclusion

- Studies and commentaries published in peer reviewed journals: no restrictions based on study design (quantitative, qualitative, case studies and RCTs)
- Official reports and guidelines.

Exclusion

- Conference abstracts
- Grey literature such as personal blogs, commentaries from relatives/carers, or news articles
- Information on non-institutional or non-NGO websites

Context.

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Give summary details of the setting or other relevant characteristics, which help define the inclusion or exclusion criteria.
This review will focus on Intensive Care Units, A&E, Acute Medical Units and related wards during COVID-19.

24. * Main outcome(s).

Give the pre-specified main (most important) outcomes of the review, including details of how the outcome is defined and measured and when these measurements are made, if these are part of the review inclusion criteria.
This review will provide a detailed understanding of the characteristics of redeployment of healthcare professionals during COVID-19. Potential outcomes will include:

- Principles and key objectives of deployment/redeployment of healthcare staff during COVID-19 (i.e. NHS guidelines for redeployment)
- Measures of staff experience and satisfaction with redeployment
- Results from pre/post tests in relation to training
- Evaluation of training programmes (i.e. Kirkpatrick model).

* Measures of effect

Please specify the effect measure(s) for you main outcome(s) e.g. relative risks, odds ratios, risk difference, and/or 'number needed to treat.
Not Applicable

25. * Additional outcome(s).

List the pre-specified additional outcomes of the review, with a similar level of detail to that required for main outcomes. Where there are no additional outcomes please state 'None' or 'Not applicable' as appropriate
to the review
Not applicable

26. * Data extraction (selection and coding).

Describe how studies will be selected for inclusion. State what data will be extracted or obtained. State how this will be done and recorded.
Search results will be deduplicated and imported into Rayyan to manage the references for the tittle and abstract stages of the screening. Two researchers will review articles independently and discuss any discrepancies until consensus is reached.
At least two review authors will extract relevant information from included studies onto a data extraction form developed on REDCap using a pre-defined list of data to extract (see below). We will resolve any disagreement by discussion with a third member of the team.
The list of data to extract will be created after an initial review of the selected articles. We will extract descriptive data and any primary qualitative data or themes identified relating to redeployment and training for redeployed healthcare staff working in emergency services during COVID-19
Preliminary list of data to extract:

- Citation details
- Location of study
- Data collection methods / source of data (for secondary data analysis studies)
- Data analysis methods
- Population:
  - Professional group
  - Sample size
  - Sociodemographic and socioeconomic characteristics of the sample (gender, age, educational level, income level, and ethnicity)
- For research question 1: redeployment
  - Redeployment process

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- Objectives
- Level at which it occurred (hospital, regional, national…)
- Implementation strategies
- Learnings – what worked

- Redeployed staff
  - Training needs
  - Obligations

- For research question 2: Training
  - Training programmes offered/developed
  - Training programme evaluations
  - Learnings – what worked

27. * Risk of bias (quality) assessment.

State which characteristics of the studies will be assessed and/or any formal risk of bias/quality assessment tools that will be used.

We expect a heterogeneous group of studies using different questions and outcomes, therefore a Mixed Methods Appraisal Tool to assess if the research question, methods, results and discussion align, alongside quality criteria specific to quantitative, qualitative and RCT studies will be applied. We will apply the AACODS checklist (Authority, Accuracy, Coverage, Objectivity, Date, Significance) to evaluate grey literature.

28. * Strategy for data synthesis.

*Describe the methods you plan to use to synthesise data. This **must not be generic text** but should be **specific to your review** and describe how the proposed approach will be applied to your data. If meta- analysis is planned, describe the models to be used, methods to explore statistical heterogeneity, and software package to be used.*

We will synthesise qualitative study outputs applying framework-based thematic analysis (Braun & Clarke 2006; Gale et al. 2013), narrative synthesis of quantitative outputs (Popay et al. 2006), and an interpretative synthesis combining both results (Barnett-Page and Thomas, 2009).

29. * Analysis of subgroups or subsets.

State any planned investigation of 'subgroups'. Be clear and specific about which type of study or participant will be included in each group or covariate investigated. State the planned analytic approach.

We do not anticipate it will be possible to conduct subgroup analysis. As part of our synthesis of results we will list differences between professional groups and locations.

Dissemination plans.

Do you intend to publish the review on completion?

Peer reviewed publication

Blog article

Supplementary Material 2: Search strategy

Ovid: Medline, Psychinfo, Embase, and HMIC:

1: "Intensive care" OR "acute medical units" OR "critical care" OR "critically ill" OR "critical illness" OR "intensive treatment unit" OR "intensive therapy unit" OR "high dependency unit" or ICU or ITU or HDU

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2. Training OR education OR course OR prepar* OR "clinical competence" OR "education intervention" OR "training intervention" OR Skill OR "competency-based education" OR "simulation" OR "simulation training"
3. Redeploy* OR deploy* OR reallocation OR reallocate
4. staff OR "Healthcare worker" OR HCW OR "healthcare professional" OR clinician OR nurse OR anaesthetist OR doctor OR trainee OR AHP OR allied health professional

Search 1 AND 2 AND 3 AND 4
Time limit Dec 2019-Dec 2020
Time limit Dec 2020-July 2021

MedRxiv

Redeploy* AND Covid-19

Web of Science

1: "Intensive care" OR "acute medical units" OR "critical care" OR "critically ill" OR "critical illness" OR "intensive treatment unit" OR "intensive therapy unit" OR "high dependency unit" or ICU or ITU or HDU
2. Training OR education OR course OR prepar* OR "clinical competence" OR "education intervention" OR "training intervention" OR Skill OR "competency-based education" OR "simulation" OR "simulation training"
3. Redeploy* OR deploy* OR reallocation OR reallocate
4. staff OR "Healthcare worker" OR HCW OR "healthcare professional" OR clinician OR nurse OR anaesthetist OR doctor OR trainee OR AHP OR allied health professional

Search 1 AND 2 AND 3 AND 4
Time limit to “last five years”

OpenGrey and SSRN

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Redeploy* AND Covid-19
"coronavirus"

TRIP
"intensive care" AND training AND redeploy*

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5 (see supplementary material 1)
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5 (see supplementary material 2)
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5-6
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	5-6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	5-6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	N/A

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PRISMA 2009 Checklist

Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	N/A
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	N/A
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	6-7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	8-9
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	N/A
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	N/A
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	N/A
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	N/A
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	14-15
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	15
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	14-16
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	2

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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For more information, visit: **www.prisma-statement.org**.

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